DRAFT — UNOFFICIAL — NOT FOR OPERATIONAL USE

PUBLICATION

DEPENDENCY-MAP

USAREUR-AF

MAVEN TRAINING CORPUS — DEPENDENCY MAP

USAREUR-AF Operational Data Team | C2DAO Training Branch

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

DRAFT — NOT FOR OFFICIAL USE. FOR TRAINING PLANNING PURPOSES ONLY.

24 MARCH 2026

DRAFT — UNOFFICIAL — NOT FOR OPERATIONAL USE

MAVEN TRAINING CORPUS — DEPENDENCY MAP

USAREUR-AF OPERATIONAL DATA TEAM | C2DAO TRAINING BRANCH

PUBLISHED OUTPUT: The authoritative HTML dependency map is `maven_training/DEPENDENCY_MAP.html` (7,400+ lines). **Do NOT overwrite it with the short ~900-line file produced by `scripts/generate_dep_map.py`.** That script generates a disposable summary — it must NEVER write to `DEPENDENCY_MAP.html`. If you need to update the dep map, edit `DEPENDENCY_MAP.html` directly. Never open, reference, or show the short generated version.

Generated: 13 March 2026 **Last structural change:** 18 March 2026 — Added T3 Train-the-Trainer tracks: T3-I (Instructor Certification, 5 days, prereq TM-30) and T3-F (MSC Force Multiplier / Unit Data Trainer, half day, prereq TM-20); added TM-40N (UI/UX Designer) and TM-40O (Platform Engineer) specialist tracks with TM-50N/TM-50O advanced tracks; ASF role alignment (PM, Designer, SWE, Platform Engineer); prereq chain extended G–O **Last updated:** 18 March 2026 (integrated USAREUR-AF Data and Analytics Strategy (CG-signed May 2025) and Unified Data Transition Strategy (CUI quarterly product cycle) into corpus — Sec 0B, Sec 5, External Refs, doctrine panel, publications index; changelog #118–119) **Audit method:** Full manual read of all source files across 6 parallel agent passes (13 March); ralf branch content migration (14 March); Palantir Developers video mapping (14 March); full ralf file inventory and PDF build (14 March); full content read of all 30 CDA slide decks (14 March) — level assignments corrected corpus-wide; pre-v2.0 content accuracy and consistency audit across all curriculum, app, widget, and PPTX files (14 March) **Status at generation:** Corpus at v2.0-ready state. All naming and content issues resolved. TM-40D/E/F prereq discrepancy resolved 2026-03-14 — all WFF tracks (TM-40A–F) now require TM-30 as terminal prerequisite. TM-40N (UI/UX Designer), TM-40O (Platform Engineer), and corresponding TM-50N/TM-50O advanced tracks added 2026-03-18 with ASF role alignment. Pre-v2.0 audit complete: 8 content fixes applied (README.md, POLICY_LETTER.md, cheatsheet.md, Home.tsx, TM40.tsx, mss_info_app/index.html, .gitignore, MSS_Project_Overview.pptx); MSS_Training_Progression.pptx specialist column fully rebuilt with clean WFF (A–F) / Specialist (G–O) grouping. PDFs regenerated (3 source-changed files rebuilt). See changelog entry #95–103.

READING THIS MAP

Relationship symbols: - → requires / directly depends on - ↔ companion / integration reference (not a prereq) - ⇢ recommended reading for

Publication type tags:

Tag	Type	Description
[COURSE]	Course & Assessment	TM (Technical Manual), Syllabus, Exercise, Exam, Concepts Guide — anything a student completes or is evaluated against
[PPT]	Slide Deck	CDA slide deck (conceptual prereq; `maven_training/source_material/course_portal/`)
[REF]	Reference	Quick-reference tools, cheatsheets, job aids — consulted during work, not studied (`maven_training/quick_reference/`)
[SUPP]	Supplemental	Supporting doctrine, guidance, or architecture docs that enrich understanding but are not part of the course sequence (`maven_training/doctrine/`, `source_material/cda_docs/`)
[EA]	Enterprise Architecture	EA series docs (`maven_training/doctrine/enterprise_architecture/`)
[CANON]	CDA Canon	CDA canon and doctrine docs (`maven_training/doctrine/cda_doctrine/`)

PREREQUISITE CHAIN (AUTHORITATIVE)

```
[TM-SL] Senior Leader ← O-5 / E-9+; replaces TM-10 for SL; terminal (no further
progression)
        |                No prereqs. No TM credit granted. Does not feed into TM
pipeline.
        |
TM-10 (all other personnel)
  └── TM-20 (builders)
        |
        ├── [FBC] Foundry Bootcamp ← parallel track; outside TM chain; no TM credit
granted
        |         Prereq: TM-20 Go + command-validated project. See
foundry_bootcamp/FBC_GUIDE.md
        |
        ├── [T3-F] MSC Force Multiplier ← Unit Data Trainer certification (half day)
        |         Prereq: TM-20 Go + commander nomination. Authorizes: TM-10 delivery,
        |         TM-10 exam proctoring, training status reporting to C2DAO.
        |         Does NOT grant TM-30 credit. Does NOT unlock TM-40 enrollment.
        |
        └── TM-30 (advanced builders / data-adjacent)
              |
              ├── [T3-I] Instructor Certification ← C2DAO instructor pipeline
              |         Prereq: TM-30 Go + C2DAO Training OIC selection.
              |         Phase 1: 5-day classroom. Phase 2: supervised practicum
```

```
(separate).
              │         Graduates authorized to teach TM-10 through TM-30 (+ domain
tracks per FDP).
              │
              ├── TM-40A (Intelligence WFF)
              ├── TM-40B (Fires WFF)
              ├── TM-40C (Movement & Maneuver WFF)
              ├── TM-40D (Sustainment WFF)
              ├── TM-40E (Protection WFF)
              ├── TM-40F (Mission Command WFF)
              ├── TM-40G (ORSA) ──────────────→ TM-50G (Advanced ORSA)
              ├── TM-40H (AI Engineer) ─────────→ TM-50H (Advanced AI Eng)
              ├── TM-40M (ML Engineer) ─────────→ TM-50M (Advanced ML Eng)
              ├── TM-40J (Program Manager) ─────→ TM-50J (Advanced PM)
              ├── TM-40K (Knowledge Manager) ──→ TM-50K (Advanced KM)
              ├── TM-40L (Software Engineer) ──→ TM-50L (Advanced SWE)
              ├── TM-40N (UI/UX Designer) ─────→ TM-50N (Advanced UI/UX)
              └── TM-40O (Platform Engineer) ──→ TM-50O (Advanced Platform Eng)
```

NOTE

TM-50A through TM-50F do NOT exist. Only TM-50G–O (advanced specialist) are valid. WFF tracks (TM-40A–F) require TM-30 (same prereq chain as Specialist tracks) — design decision 2026-03-13. Specialist tracks (TM-40G–O) all require TM-30 (required, not recommended). TM-40N (UI/UX Designer) and TM-40O (Platform Engineer) align with Army Software Factory (ASF) role tracks. Together with TM-40J (PM) and TM-40L (SWE), these four tracks cover all ASF career paths. **Cross-track companions:** TM-40N ↔ TM-40J ↔ TM-40L form the ASF "balanced team" triad (Designer + PM + Engineer). TM-40O ↔ TM-40L (platform ↔ application boundary). TM-40O ↔ TM-40H (infrastructure for AI workloads). **Foundry Bootcamp (FBC)** is a separate quarterly event outside the TM chain. Prereq: TM-20. Does NOT grant TM-30 credit. Does NOT unlock TM-40 enrollment. See training_management/FOUNDRY_BOOTCAMP_SOP.md. **Senior Leader (TM-SL)** is a separate 1-day executive course outside the TM chain. Audience: O-5 / E-9+. Replaces TM-10 for senior leaders. Terminal — does not unlock TM-20 or any further progression. Does NOT grant TM-10 credit. **T3-F (MSC Force Multiplier)** certifies Unit Data Trainers (UDTs) at the MSC level. Prereq: TM-20. Half day. Authorizes TM-10 delivery, TM-10 exam proctoring, and training status reporting. UDTs are the forward training presence between MTT rotations. Does NOT authorize TM-20 or any TM-30+ delivery. **T3-I (Instructor Certification)** is the formal C2DAO instructor pipeline. Prereq: TM-30 + OIC selection. Phase 1 is a 5-day classroom course (adult learning, lab facilitation, evaluation calibration, microteaching). Phase 2 is a supervised practicum scheduled around actual course iterations. Graduates are authorized to teach independently per domain qualification (FDP §2-1). See UNIT_DATA_TRAINER_SOP.md for T3-F governance and FACULTY_DEVELOPMENT_PLAN.md for T3-I certification standards.

CDA SLIDE LIBRARY OVERLAY (CONCEPTUAL PREREQS)

The CDA slide decks provide conceptual grounding that the TM series assumes but does not teach. They are recommended reading, not required prerequisites. They feed into the TM chain as follows.

Last alignment audit: 14 March 2026 — full content read of all 30 decks; level assignments corrected based on actual slide content vs. TM audience (user / no-code builder / advanced builder / code-level).

```
CDA "TM-10" (2 decks) ──────────────────⇢ TM-10 / all personnel
  army_data_orientation_v1             ⇢ TM-10 — WHY MSS exists; Army data landscape;
all personnel
                                           v2 (army_data_orientation_v2.pptx) patches
slides 12–14
                                           for end-user (consumer) audience; v1 had
builder framing
  mss_platform_overview                ⇢ TM-10 — NEW (2026-03-14): HOW to use MSS as
an end user
                                           Covers: platform navigation, reading
dashboards, AIP
                                           queries, submitting forms, classification
handling,
                                           problem reporting, user vs. builder
boundary
                                           File:
source_material/course_portal/downloads/mss_platform_overview.pptx

CDA "Intro To Data" (10 decks) ────────⇢ TM-20 (conceptual prereqs for no-code builders)
  architecture_primer                  ⇢ before TM-20 (stability stack, activities vs
enablers)
                                           [was tagged TM-10 — wrong; no end-user
content]
  2026_Data_Stack_Deep_Dive            ⇢ before TM-20 (five-layer L1–L5 stack
overview)
  AI_ML_Beyond_The_Hype                ⇢ TM-20 (what AI actually is; why clean ontology
enables AIP)
                                           [was tagged TM-40H — wrong; no code
content; TM-20 builders
                                           need this to understand why good data
enables AIP]
  The_Semantic_Layer_Instructions      ⇢ TM-20 (slides 1–6) / TM-30 (slides 7–11
governance content)
  What_Is_An_Ontology                  ⇢ TM-20 (three building blocks, definition
clarity test)
  L5_Activation_Interface_Deep_Dive    ⇢ TM-20 (Workshop/activation decision framework
slides 1–5)
                                           NOTE: slides 6–9 (Custom React Widgets,
Streamlit, Plotly)
                                           are TM-40L content — not TM-20
  Dont_Filter_This_Isnt_Excel          ⇢ TM-20 (platform mindset shift; essential for
all builders)
  L2_Ingestion_Integration_Deep_Dive   ⇢ TM-30 (CDC, DAGs, orchestration, schema
```

```
evolution)
                                                [was tagged TM-20 — wrong; no-code builders
don't design CDC]
  Data_Modeling_Foundations            ⇢ TM-30 (Kimball, Inmon, DataVault, Enterprise
Compass)
                                                [was tagged TM-20 — wrong; too academic for
no-code builders]
  Data_Platforms_Cloud_Deep_Dive       ⇢ TM-30 (JWCC, cross-domain, edge, OCONUS
design concerns)
                                                [was tagged TM-20 — slides 1–4 fine at TM-
20 but 5–10 are TM-30]
  decision_advantage_deep_dive         ⇢ TM-30 / all WFF tracks TM-40A–F (ADP 3-13
doctrine framework;
                                                decision-as-node-graph; maps to Foundry
object types)
                                                [was tagged TM-40A — wrong; not Intel-WFF-
specific; no code]

CDA "Data 101" (6 decks) ──────────────⇢ TM-20 / TM-30
  The_Four_Layers                      ⇢ TM-20 (ontology vs dataset vs pipeline vs
app; essential)
  Data_Modeling_Fundamentals_Level1    ⇢ TM-20 (tables, keys, cardinality, schemas,
data types)
  ObjectType_WhatToWatchFor            ⇢ TM-20 (filter-as-type anti-pattern;
downstream avalanche)
  Links_and_Relationships              ⇢ TM-20 (cardinality, direction, optionality)
  Controlled_Vocabularies              ⇢ TM-20 / TM-30 (needed before building first
object type)
                                                [was tagged TM-30 only — should start at
TM-20]
  Identity_Who_Owns_The_Key            ⇢ TM-20 / TM-30 (primary key authority, six
rules, entity resolution)
                                                [was tagged TM-30 only — fundamentals
needed at TM-20]

CDA "Data 201" (10 decks) ──────────────⇢ TM-30 / TM-40G–O
  Semantic_Modelling_Course_Intro      ⇢ TM-30 / TM-40L (entry to 12-deck series;
prereq: DM L2)
  Deck_01_Architecture_First_Principles ⇢ TM-30 / TM-40L (separation of concerns,
stability stack)
  Deck_02_Scope_Engineering            ⇢ TM-30 (schema fit vs scope fit, fit
scorecard)
  Deck_03_RDF_OWL_Foundations          ⇢ TM-30 (part of 12-deck series;
RDF/OWL/SKOS/SHACL)
                                                [was tagged TM-40K — wrong; this is Deck 3
of the same
                                                series as Decks 1, 2, 4, 5, 6 which are all
TM-30]
  Deck_04_Object_Type_Varieties        ⇢ TM-30 / TM-40L (nine canonical varieties,
decision tree)
  Deck_05_Identity_Governance          ⇢ TM-30 / TM-40K (identity authority matrix,
entity resolution arch)
  Deck_06_Relationship_Modeling        ⇢ TM-30 / TM-40L (reification, role patterns,
SHACL constraints)
  Deck_07_Temporal_Bitemporal          ⇢ TM-30 (part of 12-deck series; event/state,
```

```
bitemporal patterns)
                                                [was tagged TM-40G — wrong; temporal
modeling is general TM-30;
                                                applies across all specializations not just
ORSA]
  Data_Modeling_Fundamentals_Level2    ⇢ TM-40 general / TM-40G / TM-40M / TM-40L
                                                (Kimball/Inmon/DataVault comparison,
Enterprise Compass)
  Deck_12_Capstone_Foundry             ⇢ TM-40 general (end-to-end ontology
engineering capstone;
                                                prereq: Decks 1–11; bridges TM-30 theory →
TM-40 practice)
                                                [was tagged TM-40G only — not ORSA-
specific]
```

SECTION 0B — CDA DOCTRINE AND REFERENCE MATERIALS

Enterprise Architecture (TM-30 / TM-40K / TM-40L)

File	Type	Supports
doctrine/enterprise_architecture/EA_00_REFERENCE_CARD.md	[EA]	→ TM-30, TM-40K, TM-40L
doctrine/enterprise_architecture/EA_01_FOUNDATION.md	[EA]	→ TM-30 (architecture thinking)
doctrine/enterprise_architecture/EA_02_SCHOOLS_OF_THOUGHT.md	[EA]	→ TM-30, TM-40K
doctrine/enterprise_architecture/EA_03_ARTIFACTS_AND_VIEWS.md	[EA]	→ TM-40K, TM-40L
doctrine/enterprise_architecture/EA_04_GOVERNANCE.md	[EA]	→ TM-40K
doctrine/enterprise_architecture/EA_05_MILITARY_APPLICATION.md	[EA]	→ TM-30, TM-40G–O

Command Strategy

File	Type	Supports
doctrine/USAREUR_AF_DATA_ANALYTICS_STRATEGY.md	[SUPP]	→ All TMs, COMMANDERS_GUIDE, POLICY_LETTER (USAREUR-AF Data and Analytics Strategy; CG-signed May 2025; 4 strategic outcomes, VAULTIS framework, Cognitive Hierarchy)

File	Type	Supports
doctrine/UNIFIED_DATA_TRANSITION_STRATEGY.md	[SUPP]	→ TM-30, TM-40G–O, TM-50G–O (Unified Data Transition Strategy quarterly product cycle; CUI; Discovery & Framing → PSB → Iteration & Implementation)

Senior Leader Guidance

File	Type	Supports
doctrine/CG_GUIDANCE.md	[REF]	→ TM-10, TM-20 (CG public doctrine; AUSA 2025, Green Notebook 2026)

CDA Doctrine (TM-40G–O)

File	Type	Supports
doctrine/CDA_CONSTRAINTS_AND_DIRECTIVES.md	[CNCPT]	→ TM-40G–O (12 operational constraints and directives governing CDA work)
doctrine/cda_doctrine/CDA_OVERVIEW.md	[CNCPT]	→ TM-30 (bridge: CDA user taxonomy, enterprise index)
doctrine/cda_doctrine/CDA_DOCTRINE_OVERVIEW.md	[CNCPT]	→ TM-40G–O (doctrine-driven development)
doctrine/cda_doctrine/CDA_DOCTRINE_AGENT.md	[CNCPT]	→ TM-40H, TM-40L
doctrine/cda_doctrine/CDA_AVT25_ASSESSMENT.md	[CNCPT]	→ TM-40G, TM-40H
doctrine/cda_doctrine/CDA_IDENTITY_VS_CLASSIFICATION.md	[CNCPT]	→ TM-30, TM-40K, TM-40L
doctrine/ONTOLOGY_DESIGN_PRINCIPLES.md	[CNCPT]	→ TM-40H, TM-40K, TM-40L, TM-50H/K/L (DDD, DRY, Open/Closed, PECS, Composition; Foundry-specific)

CDA Architecture Doctrine — Agents Series (TM-40H/M/K/L, TM-50H/M/K/L)

Bedrock → specialization hierarchy. Core Principles govern; specialized docs extend. | File | Type | Supports | |---|---|---| | doctrine/cda_doctrine/agents/CDA_AGENTS_OVERVIEW.md | [CNCPT] | → index for agents subdirectory; hierarchy diagram | | doctrine/cda_doctrine/agents/CDA_AGENTS_CORE_PRINCIPLES.md | [CNCPT] | → TM-40H, TM-40M, TM-40K, TM-40L, TM-50H/M/K/L (12 bedrock principles: Stability Stack, Four-Layer Architecture, Scope

Engineering, Nine Object Types, Identity Governance, VAULTIS-A, 14 anti-patterns) | | doctrine/cda_doctrine/agents/CDA_AGENTS_ONTOLOGY_ENGINEER.md | [CNCPT] | → TM-40H, TM-40K, TM-40L, TM-50H/K/L (specializes Core Principles 3/7/8/9/10/11; RDF, OWL2, SHACL, SPARQL, PROV-O; SHACL shapes, competency questions) | | doctrine/cda_doctrine/agents/CDA_AGENTS_ENTITY_RESOLUTION.md | [CNCPT] | → TM-40H, TM-40M, TM-40L, TM-50H/M/L (9-stage ER pipeline; blocking, scoring, decisioning, cluster building, survivorship; threshold bands; 10 anti-patterns) | | doctrine/cda_doctrine/agents/CDA_AGENTS_INGESTION_INTEGRATION.md | [CNCPT] | → TM-40H, TM-40M, TM-40L, TM-50H/M/L (Five-Stage Ingestion Pattern; ELT over ETL; Dagster DAG; VAULTIS-A per pipeline; 9 anti-patterns; 10-item output checklist) |

GDAP — Global Doctrine Alignment Platform (TM-40H/M/K/L, TM-50H/M/K/L)

File	Type	Supports
doctrine/gdap/GDAP_OVERVIEW.md	[CNCPT]	→ TM-40H, TM-40M, TM-40K, TM-40L (platform quickstart, LlamaIndex retrieval stack, API endpoints)
doctrine/gdap/GDAP_VISION.md	[CNCPT]	→ TM-40H, TM-40M, TM-40K, TM-40L, TM-50H/M/K/L (full vision: 10 use-case domains, 20 pipeline steps, DVEE, coalition seam detection)
doctrine/gdap/GDAP_PERSISTENCE_STRATEGY.md	[CNCPT]	→ TM-40H, TM-40M, TM-40L, TM-50H/M/L (DuckDB = canonical truth; LlamaIndex = derived serving artifacts; versioning/rebuild/rollback policy)
doctrine/gdap/GDAP_ACCEPTANCE_TESTS.md	[CNCPT]	→ TM-40H, TM-40M, TM-40L, TM-50H/M/L (P0/P1/P2 release gates across 9 sections: A=Foundation, B=Ingestion, C=Retrieval, D=Routing, E=Response Quality, F=Agents, G=Evaluation, H=Ops, I=Consumer Delight)
doctrine/gdap/GDAP_ADR_0001_LLAMAINDEX.md	[CNCPT]	→ TM-40H, TM-40M, TM-40L, TM-50H/M/L (ADR-0001 accepted: DuckDB owns canonical truth; LlamaIndex owns serving artifacts; boundary rules)

MIM — MIP Information Model (TM-40H/M/K/L, TM-50H/M/K/L)

File	Type	Supports
doctrine/mim/MIM_OVERVIEW.md	[CNCPT]	→ TM-40H, TM-40M, TM-40K, TM-40L (toolchain: HTML/XSD parsers, Foundry backend, repo structure, package quick ref)
doctrine/mim/MIM_STANDARD.md	[CNCPT]	→ TM-40H, TM-40M, TM-40K, TM-40L, TM-50H/M/K/L (MIM semantic model: namespaces, type/instance/status, roles, discriminators, code types, design principles, Semantic IDs)

File	Type	Supports
doctrine/mim/MIM_STATE.md	[CNCPT]	→ TM-40H, TM-40M, TM-40L, TM-50H/M/L (project snapshot 2026-02-28: maturity by area, passing/failing tests, key gaps, 30/60/90 priorities)
doctrine/mim/MIM_ACADEMICS.md	[CNCPT]	→ TM-40H, TM-40M, TM-40K, TM-40L, TM-50H/M/K/L (Dr. Gerz / NATO interoperability; MIM↔Foundry structural alignment analysis; UML↔OT mapping; MIP governance)
doctrine/mim/MIM_DECISION_RECORDS.md	[CNCPT]	→ TM-40H, TM-40L (ADR structure decision: repo-level and per-package ADRs)
doctrine/mim/MIM_FUTURE_CLASSES.md	[CNCPT]	→ TM-40H, TM-40M, TM-40L, TM-50H/M/L (planned: mim-studio, mim-ui, adapters, backends, mim-sdk, mim-provenance)
doctrine/mim/MIM_ONTOLOGY_DOCS.md	[CNCPT]	→ TM-40H, TM-40L, TM-50H/L (OSDK Maker Package full TypeScript API: SPTs, Value Types, Interfaces, Objects, Links, Interface Link Constraints, Actions, Derived Properties)

Canon Reference (TM-40G–O)

File	Type	Supports
doctrine/cda_doctrine/canon/CANON_ADP_CROSSWALK.md	[CANON]	→ all specialist tracks TM-40G–O
doctrine/cda_doctrine/canon/CANON_CONDITIONS_INDICATORS_THRESHOLDS.md	[CANON]	→ TM-40G, TM-40H
doctrine/cda_doctrine/canon/CANON_ENGAGEMENT.md	[CANON]	→ TM-40G, TM-40H
doctrine/cda_doctrine/canon/CANON_INFORMATION.md	[CANON]	→ TM-40H, TM-40K

Quick Reference

File	Type	Supports
quick_reference/cda_reference/EA_VS_DA.md	[REF]	→ TM-30, TM-40K
quick_reference/cda_reference/ENTERPRISE_DATA_COMPASS.md	[REF]	→ TM-40J, TM-40K

DRAFT

File	Type	Supports
quick_reference/cda_reference/LESSONS_LEARNED.md	[REF]	→ all tracks
quick_reference/cda_reference/PLAN_FOR_SUCCESS.md	[REF]	→ TM-40J, TM-30
quick_reference/cda_reference/DOCTRINE_ELEMENT_FOUNDRY_MAPPING.md	[REF]	→ TM-40H, TM-40M, TM-40L, TM-50H/M/L (DoctrineElement property map, bi-temporal fields, TypeScript interface, OSDK query patterns, link types, enums)
exercises/exams/EXAM_CDA_FINAL.md	[EXAM]	→ assesses Intro To Data + Data 101 + Data 201

Interactive Apps

App	Type	Supports
source_material/cda_apps/ea-vs-da-reference/	[REF]	→ TM-30, TM-40K, TM-40L
source_material/cda_apps/enterprise-data-compass/	[REF]	→ TM-40J, TM-40K
source_material/cda_apps/lessons-learned/	[REF]	→ all tracks
source_material/cda_apps/plan-for-success/	[REF]	→ TM-40J, TM-30
source_material/course_portal/	[PPT]	→ see CDA Slide Library above

SECTION 0C — SENIOR LEADER EXECUTIVE COURSE (TM-SL)

Audience: O-5 / E-9+. No prereqs. Terminal qualification — does not feed into TM pipeline.

Replaces TM-10 for senior leaders. Does NOT grant TM-10 credit.

File	Type	Outbound Dependencies
TM_SL_SENIOR_LEADER.md	[TM]	⇢ CG_GUIDANCE (strategic context), ⇢ DATA_LITERACY_technical_reference, ⇢ TM-10 (if SL wants hands-on)
CONCEPTS_GUIDE_TM_SL_SENIOR_LEADER.md	[CNCPT]	↔ TM_SL_SENIOR_LEADER [TM] (read-ahead companion)

File	Type	Outbound Dependencies
SYLLABUS_TM_SL.md	[SYL]	→ TM-SL [TM]
(no Exercise)	—	Assessment is organizational, not individual
(no Exam)	—	Qualification recorded as "Complete" upon attendance

Inbound references from: DEPENDENCY_MAP prereq chain, ENROLLMENT_SOP, ANNUAL_TRAINING_SCHEDULE, MTP, mss_info_app/index.html (if applicable)

Recommended pre-reading: ⇢ CG_GUIDANCE.md · DATA_LITERACY_technical_reference Ch 1

Key governance note: TM-SL is a 1-day instructor-led course for O-5 / E-9+ personnel. It replaces TM-10 for this population. It covers formation-level data product impact, agile project management, working with data professionals, governance, and the training pipeline. It is terminal — senior leaders who want hands-on qualification enroll in TM-10 separately.

SECTION 1 — BASE TRACKS

TM-10: Maven User

File	Type	Outbound Dependencies
TM_10_MAVEN_USER.md	[TM]	→ TM-20 (next step), ⇢ DATA_LITERACY_technical_reference
(no Concepts Guide)	—	—
SYLLABUS_TM10.md	[SYL]	→ TM-10 [TM]
EX_10_operator_basics/EXERCISE.md	[EX]	→ TM-10 [TM]
EXAM_TM10_PRE.md	[EXAM]	assesses TM-10 readiness
EXAM_TM10_SUPPLEMENTAL.md	[EXAM]	optional supplemental knowledge check for TM-10

Inbound references from: TM-20, TM-30, TM-40A–F (all list TM-10 as prereq), SYLLABUS_TM20, SYLLABUS_TM30, all TM-40/50 syllabi, MTP, POI, CAD, TEO, ENROLLMENT_SOP, ANNUAL_TRAINING_SCHEDULE, CHEATSHEET, README, QUICK_START

Recommended CDA decks [PPT]: ⇢ army_data_orientation_v1 · architecture_primer · 2026_Data_Stack_Deep_Dive

TM-20: Builder

File	Type	Outbound Dependencies
TM_20_BUILDER.md	[TM]	→ TM-10 (prereq), → TM-30 (escalation), ↔ TM-10 Ch 4/5/8 task refs, ⇢ TM-40M, ⇢ TM-40L
CONCEPTS_GUIDE_TM20_BUILDER.md	[CNCPT]	↔ TM_20_BUILDER [TM] (companion), ⇢ TM-30 escalation guide
SYLLABUS_TM20.md	[SYL]	→ TM-10 (prereq)
EX_20_no_code_builder/EXERCISE.md	[EX]	→ TM-20 [TM], → NAMING_AND_GOVERNANCE_STANDARDS
EXAM_TM20_PRE.md	[EXAM]	assesses TM-20 readiness
EXAM_TM20_POST.md	[EXAM]	assesses TM-20 completion

Inbound references from: TM-30, TM-40A–O (prereq chain via TM-30), all training mgmt docs, CHEATSHEET, README

Recommended CDA decks [PPT]: ⇢ What_Is_An_Ontology · Data_Modeling_Foundations · The_Semantic_Layer_Instructions · L2_Ingestion_Integration_Deep_Dive · L5_Activation_Interface_Deep_Dive · Dont_Filter_This_Isnt_Excel · The_Four_Layers · Data_Modeling_Fundamentals_Level1 · ObjectType_WhatToWatchFor · Links_and_Relationships

TM-30: Advanced Builder

File	Type	Outbound Dependencies
TM_30_ADVANCED_BUILDER.md	[TM]	→ TM-10 (prereq), → TM-20 (prereq), ⇢ DATA_LITERACY_technical_reference
CONCEPTS_GUIDE_TM30_ADVANCED_BUILDER.md	[CNCPT]	↔ TM_30_ADVANCED_BUILDER [TM] (companion), ⇢ TM-40G–O escalation guide, ⇢ EA Series, ⇢ CDA doctrine
SYLLABUS_TM30.md	[SYL]	→ TM-10, → TM-20 (prereqs)
EX_30_advanced_builder/EXERCISE.md	[EX]	→ TM-30 [TM], → NAMING_AND_GOVERNANCE_STANDARDS

DRAFT

File	Type	Outbound Dependencies
EXAM_TM30_PRE.md	[EXAM]	assesses TM-30 readiness
EXAM_TM30_POST.md	[EXAM]	assesses TM-30 completion

Inbound references from: TM-40A–F (all WFF tracks, hard prereq), TM-40G–O (all specialist tracks, hard prereq), TM-40K (pipeline builder), TM-40L (peer note), all training mgmt docs, CHEATSHEET, README

Recommended CDA decks [PPT]: ⇢ Data_Architecture_Deep_Dive · Data_Platforms_Cloud_Deep_Dive · Semantic_Modelling_Course_Intro · Deck_01_Architecture_First_Principles · Deck_02_Scope_Engineering · Deck_04_Object_Type_Varieties · Deck_05_Identity_Governance · Deck_06_Relationship_Modeling · Controlled_Vocabularies · Identity_Who_Owns_The_Key

SECTION 1B — FOUNDRY BOOTCAMP (FBC)

Prereq: TM-20 Go on file + command-approved project

NOT part of the TM-10 → TM-50 sequence; does NOT grant TM credit

Quarterly 5-day supervised build event. Does NOT grant TM-30 credit. Does NOT unlock TM-40 enrollment. See training_management/FOUNDRY_BOOTCAMP_SOP.md.

File	Type	Outbound Dependencies
foundry_bootcamp/FBC_GUIDE.md	[FBC]	→ TM-20 (prereq); does NOT → TM-30 (explicitly not a substitute)
foundry_bootcamp/ENVIRONMENT_SETUP.md	[FBC]	→ FBC_GUIDE (coordinator checklist for bootcamp setup)
foundry_bootcamp/SPRINT_PACKAGE.md	[FBC]	→ FBC_GUIDE Ch 5 (evaluator/coordinator standards); → ENROLLMENT_SOP (enrollment workflow)
training_management/FOUNDRY_BOOTCAMP_SOP.md	[SOP]	→ TM-20 (prereq); → FBC_GUIDE, → SPRINT_PACKAGE, → ENVIRONMENT_SETUP (references all FBC docs); → ANNUAL_TRAINING_SCHEDULE (quarterly calendar)

Inbound references from: DEPENDENCY_MAP prereq chain, DEMO_BRIEF_CG, WHITE_PAPER_MSS_TRAINING, mss_info_app/index.html (if applicable)

SECTION 1C — TRAIN THE TRAINER (T3)

T3-F: MSC Force Multiplier (Unit Data Trainer Certification)

Prereq: TM-20 Go on file + unit commander nomination

Half day (4 hours) — Authorizes TM-10 delivery, TM-10 exam proctoring, training status reporting

File	Type	Outbound Dependencies
tm/T3_F_msc_force_multiplier/T3_F_MSC_FORCE_MULTIPLIER.md	[COURSE]	→ TM-20 (prereq); → TM-10 (delivery authorization); ↔ INSTRUCTOR_OVERVIEW (instructor ecosystem context); → UNIT_DATA_TRAINER_SOP (governance)
tm/T3_F_msc_force_multiplier/CONCEPTS_GUIDE_T3F_MSC_FORCE_MULTIPLIER.md	[CNCPT]	→ TM-20 (prereq baseline); ⇢ T3-F [COURSE] (pre-course reading)
syllabi/SYLLABUS_T3F.md	[SYL]	→ TM-20 (prereq); → CONCEPTS_GUIDE_T3F (pre-course reading)
exercises/EX_T3F_force_multiplier/EXERCISE.md	[EX]	→ T3-F [COURSE]; → TM-10 lesson plans (teach-back source material)
exercises/EX_T3F_force_multiplier/ENVIRONMENT_SETUP.md	[EX]	→ T3-F [COURSE] (pre-exercise environment checklist)
exercises/exams/EXAM_T3F_PRE.md	[EXAM]	→ T3-F [COURSE] (diagnostic — Day 1)
exercises/exams/EXAM_T3F_POST.md	[EXAM]	→ T3-F [COURSE] (knowledge check — Day 3)
training_management/lesson_plans/T3F_LESSON_PLAN_OUTLINES.md	[LP]	→ T3-F [COURSE]; → TM-10 lesson plans
training_management/UNIT_DATA_TRAINER_SOP.md	[SOP]	→ T3-F (cert requirement); → C2DAO reporting procedures; → ENROLLMENT_SOP (unit-level enrollment workflow)

Inbound references from: DEPENDENCY_MAP prereq chain, FACULTY_DEVELOPMENT_PLAN (instructor ecosystem), INSTRUCTOR_OVERVIEW (UDT tier), TASKORD_MSS_TRAINING_CELL (MTT delivery of T3-F at MSC sites)

Key governance note: T3-F creates a sustainment capability, not a replacement for C2DAO instructors. UDTs deliver TM-10 between MTT rotations. UDTs do NOT deliver TM-20, TM-30, or any TM-40/TM-50 courses. Curriculum authority remains with C2DAO.

T3-I: Instructor Certification (C2DAO Instructor Pipeline)

Prereq: TM-30 Go on file + C2DAO Training OIC selection

Phase 1: 5 days (40 hours) classroom — Phase 2: supervised practicum (scheduled separately)

File	Type	Outbound Dependencies
tm/T3_I_instructor_certification/T3_I_INSTRUCTOR_CERTIFICATION.md	[COURSE]	→ TM-30 (prereq); → TM-10/TM-20/TM-30 lesson plans (platform walkthrough); → TEO_MSS (evaluation calibration); → FACULTY_DEVELOPMENT_PLAN (certification standards)
tm/T3_I_instructor_certification/CONCEPTS_GUIDE_T3I_INSTRUCTOR_CERTIFICATION.md	[CNCPT]	→ TM-30 (prereq baseline); ⇢ T3-I [COURSE] (pre-course reading); ⇢ TP 350-70-3, TP 350-70-7
syllabi/SYLLABUS_T3I.md	[SYL]	→ TM-30 (prereq); → CONCEPTS_GUIDE_T3I (pre-course reading); → FDP §2-1, §6-1
exercises/EX_T3I_instructor_certification/EXERCISE.md	[EX]	→ T3-I [COURSE]; → TM-10/TM-20/TM-30 lesson plans (microteaching source); → TEO_MSS (evaluation scoring)
exercises/EX_T3I_instructor_certification/ENVIRONMENT_SETUP.md	[EX]	→ T3-I [COURSE] (pre-exercise environment checklist)
exercises/exams/EXAM_T3I_PRE.md	[EXAM]	→ T3-I [COURSE] (diagnostic — Day 1)
exercises/exams/EXAM_T3I_POST.md	[EXAM]	→ T3-I [COURSE] (knowledge check — Day 5)
training_management/lesson_plans/T3I_LESSON_PLAN_OUTLINES.md	[LP]	→ T3-I [COURSE]; → all TM-10/TM-20/TM-30 materials

Inbound references from: DEPENDENCY_MAP prereq chain, FACULTY_DEVELOPMENT_PLAN (certification phases), INSTRUCTOR_OVERVIEW (instructor tier structure), POI_MSS (program scope)

Key governance note: T3-I replaces the previous ad-hoc 4-phase apprenticeship with a structured, documented certification pathway. Phase 1 (classroom) covers adult learning principles, platform deep-dive from the instructor seat, lab facilitation, Go/No-Go standardization, microteaching, and common trainee error management. Phase 2 (supervised practicum) is scheduled around actual course iterations — the candidate co-teaches, then leads under observation. Domain qualification for TM-40/TM-50 delivery is a separate requirement per FDP §2-1.

T3 cross-references: - T3-F ↔ T3-I: T3-I graduates may deliver T3-F during MTT rotations. T3-F is introduced in T3-I Ch 1 as the unit-level tier. - T3-F → UNIT_DATA_TRAINER_SOP: Governs UDT authorities, limitations, reporting, and recertification. - T3-I → FACULTY_DEVELOPMENT_PLAN: Defines instructor tiers (Instructor → Senior Instructor → Master Instructor), certification phases, and recertification requirements.

SECTION 2 — WFF FUNCTIONAL TRACKS (TM-40A–F)

Prereq for all: TM-10 + TM-20 + TM-30 + track Concepts Guide

TM-40A: Intelligence

File	Type	Outbound Dependencies
TM_40A_INTELLIGENCE.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40A (prereq read), ↔ TM-40F (CCIR coordination)
CONCEPTS_GUIDE_TM40A_INTELLIGENCE.md	[CNCPT]	→ TM-10 (prereq baseline), ↔ TM-40A [TM] (Appendix D), ⇢ FM 2-0, ATP 2-01.3, ATP 2-01
SYLLABUS_TM40A.md	[SYL]	→ TM-10, → TM-20, → TM-30, ⇢ FM 2-0, ATP 2-01
EX_40A_intelligence/EXERCISE.md	[EX]	→ TM-40A [TM]
EXAM_TM40A_PRE.md	[EXAM]	assesses TM-40A readiness
EXAM_TM40A_POST.md	[EXAM]	assesses TM-40A completion

Inbound references from: TM-40D (companion cross-ref), TM-40E (AT intel integration), DATA_LITERACY_technical_reference, CHEATSHEET, mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ decision_advantage_deep_dive

TM-40B: Fires

File	Type	Outbound Dependencies
TM_40B_FIRES.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40B (prereq read), ⇢ TM-40L (system admin)
CONCEPTS_GUIDE_TM40B_FIRES.md	[CNCPT]	↔ TM-40B [TM] (read before TM-40B), ⇢ ADP 3-19, FM 3-09, FM 3-60, ATP 3-01.8, ATP 3-52.2, ATP 3-09.42, ATP 3-09.50
SYLLABUS_TM40B.md	[SYL]	→ TM-10, → TM-20, → TM-30, ⇢ FM 3-09, ATP 3-09.42, FM 3-60
EX_40B_fires/EXERCISE.md	[EX]	→ TM-40B [TM]
EXAM_TM40B_PRE.md	[EXAM]	assesses TM-40B readiness
EXAM_TM40B_POST.md	[EXAM]	assesses TM-40B completion

Inbound references from: TM-40D (ammo mgmt companion), TM-40E (AMD coordination), DATA_LITERACY_technical_reference, CHEATSHEET, mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ decision_advantage_deep_dive

TM-40C: Movement & Maneuver

File	Type	Outbound Dependencies
TM_40C_MOVEMENT_MANEUVER.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40C (prereq read), ↔ TM-40E (NBC/protection layer)
CONCEPTS_GUIDE_TM40C_MOVEMENT_MANEUVER.md	[CNCPT]	↔ TM-40C [TM] (doctrinal framework), ⇢ ADP 3-0, FM 3-96
SYLLABUS_TM40C.md	[SYL]	→ TM-10, → TM-20, → TM-30, ⇢ ADP 3-0, FM 3-0, ATP 3-90.90
EX_40C_movement_maneuver/EXERCISE.md	[EX]	→ TM-40C [TM]
EXAM_TM40C_PRE.md	[EXAM]	assesses TM-40C readiness
EXAM_TM40C_POST.md	[EXAM]	assesses TM-40C completion

Inbound references from: TM-40D (maneuver unit readiness companion), TM-40E (physical security/base camp siting), DATA_LITERACY_technical_reference, CHEATSHEET, mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ decision_advantage_deep_dive

TM-40D: Sustainment

File	Type	Outbound Dependencies
TM_40D_SUSTAINMENT.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40D (prereq read), ↔ TM-40A (intel/threat feed), ↔ TM-40B (ammo coordination), ↔ TM-40C (maneuver readiness), ↔ TM-40E (CBRN/convoy protection), ↔ TM-40F (COP/LOGSTAT), ⇢ TM-40M (ML models)
CONCEPTS_GUIDE_TM40D_SUSTAINMENT.md	[CNCPT]	↔ TM-40D [TM] (para 4-7), ↔ TM-40F (COP framework/S4 producer)
SYLLABUS_TM40D.md	[SYL]	→ TM-10, → TM-20, → TM-30, ⇢ ADP 4-0, FM 4-0, ATP 4-0.1
EX_40D_sustainment/EXERCISE.md	[EX]	→ TM-40D [TM]
EXAM_TM40D_PRE.md	[EXAM]	assesses TM-40D readiness
EXAM_TM40D_POST.md	[EXAM]	assesses TM-40D completion

Inbound references from: TM-40E (CBRN resupply), CHEATSHEET, mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ decision_advantage_deep_dive

TM-40E: Protection

File	Type	Outbound Dependencies
TM_40E_PROTECTION.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40E (prereq read), ↔ TM-40A (AT intel), ↔ TM-40B (AMD coordination), ↔ TM-40C (base camp siting),

File	Type	Outbound Dependencies
		↔ TM-40D (CBRN resupply/medical), ↔ TM-40F (COP/CCIR protection data), ⇢ TM-40L (pipeline construction)
CONCEPTS_GUIDE_TM40E_PROTECTION.md	[CNCPT]	↔ TM-40E [TM] (Appendix A naming, Appendix H data quality)
SYLLABUS_TM40E.md	[SYL]	→ TM-10, → TM-20, → TM-30, ⇢ ADP 3-37, ATP 3-37.2
EX_40E_protection/EXERCISE.md	[EX]	→ TM-40E [TM]
EXAM_TM40E_PRE.md	[EXAM]	assesses TM-40E readiness
EXAM_TM40E_POST.md	[EXAM]	assesses TM-40E completion

Inbound references from: TM-40C (NBC data integration), DATA_LITERACY_technical_reference, CHEATSHEET, mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ decision_advantage_deep_dive

TM-40F: Mission Command

File	Type	Outbound Dependencies
TM_40F_MISSION_COMMAND.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40F (prereq read), ↔ TM-40G (ORSA products for G35), ↔ TM-40H (AI-enabled CCIR alerts), ↔ TM-40M (predictive products), ↔ TM-40J (PM readiness/portfolio), ↔ TM-40K (KM info products/lessons learned)
CONCEPTS_GUIDE_TM40F_MISSION_COMMAND.md	[CNCPT]	↔ TM-40F [TM] (proceed to TM-40F for task instruction), → TM-10 (baseline proficiency)
SYLLABUS_TM40F.md	[SYL]	→ TM-10, → TM-20, → TM-30, ⇢ ADP 6-0, FM 6-0

File	Type	Outbound Dependencies
EX_40F_mission_command/EXERCISE.md	[EX]	→ TM-40F [TM]
EXAM_TM40F_PRE.md	[EXAM]	assesses TM-40F readiness
EXAM_TM40F_POST.md	[EXAM]	assesses TM-40F completion

Inbound references from: TM-40A (CCIR coordination), TM-40C (S3 coordination), TM-40D (COP/LOGSTAT), TM-40E (COP/CCIR protection), CONCEPTS_GUIDE_TM40D (COP framework), DATA_LITERACY_technical_reference, CHEATSHEET, mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ decision_advantage_deep_dive

SECTION 3 — SPECIALIST TRACKS (TM-40G–O)

Prereq for all: TM-10 + TM-20 + TM-30 (all required)

TM-40G: ORSA

File	Type	Outbound Dependencies
TM_40G_ORSA.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40G, ⇢ TM-40H (AI library install), ⇢ TM-40M (ML integration)
CONCEPTS_GUIDE_TM40G_ORSA.md	[CNCPT]	↔ TM-40G [TM] (companion), ⇢ TM-40G Appendix D checklist
SYLLABUS_TM40G.md	[SYL]	→ TM-10, → TM-20, → TM-30
EX_40G_orsa/EXERCISE.md	[EX]	→ TM-40G [TM]
EXAM_TM40G_PRE.md	[EXAM]	assesses TM-40G readiness
EXAM_TM40G_POST.md	[EXAM]	assesses TM-40G completion

Advanced track: → TM-50G **Inbound references from:** TM-40J (portfolio analytics), TM-50G (prereq), DATA_LITERACY_technical_reference, CHEATSHEET, task_index.html (17 tasks), mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ Data_Modeling_Fundamentals_Level2 · Deck_07_Temporal_Bitemporal · Deck_12_Capstone_Foundry

TM-40H: AI Engineer

File	Type	Outbound Dependencies
TM_40H_AI_ENGINEER.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40H, ⇢ TM-40M (fine-tuning/model topics)
CONCEPTS_GUIDE_TM40H_AI_ENGINEER.md	[CNCPT]	role distinction table with TM-40M, TM-40L; ↔ TM-40H [TM]
SYLLABUS_TM40H.md	[SYL]	→ TM-10, → TM-20, → TM-30
EX_40H_ai_engineer/EXERCISE.md	[EX]	→ TM-40H [TM]
EXAM_TM40H_PRE.md	[EXAM]	assesses TM-40H readiness
EXAM_TM40H_POST.md	[EXAM]	assesses TM-40H completion

Advanced track: → TM-50H **Inbound references from:** TM-40F (AI-enabled CCIR), TM-40G (library install), TM-40K (AIP Agent Studio), TM-50G (pipeline architecture), TM-50H (prereq), TM-50L (recommended), DATA_LITERACY_technical_reference, CHEATSHEET, task_index.html (1 task), mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ AI_ML_Beyond_The_Hype · Deck_12_Capstone_Foundry

TM-40M: ML Engineer

File	Type	Outbound Dependencies
TM_40M_ML_ENGINEER.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40M, ⇢ TM-40G (RStudio/ORSA workflows), ⇢ TM-40L (VS Code)
CONCEPTS_GUIDE_TM40M_ML_ENGINEER.md	[CNCPT]	role matrix consistent with TM-40G/H/L; ↔ TM-40M [TM]

File	Type	Outbound Dependencies
SYLLABUS_TM40M.md	[SYL]	→ TM-10, → TM-20, → TM-30
EX_40M_ml_engineer/EXERCISE.md	[EX]	→ TM-40M [TM]
EXAM_TM40M_PRE.md	[EXAM]	assesses TM-40M readiness
EXAM_TM40M_POST.md	[EXAM]	assesses TM-40M completion

Advanced track: → TM-50M **Inbound references from:** TM-40F (predictive products), TM-40G (ML integration), TM-50G (ML integration), TM-50H (fine-tuning infrastructure), TM-50M (prereq), DATA_LITERACY_technical_reference, CHEATSHEET, task_index.html (21 tasks), mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ AI_ML_Beyond_The_Hype · Data_Modeling_Fundamentals_Level2 · Deck_07_Temporal_Bitemporal

---

TM-40J: Program Manager

File	Type	Outbound Dependencies
TM_40J_PROGRAM_MANAGER.md	[TM]	→ TM-10, → TM-20, → TM-30 (all required), ⇢ TM-40H, ⇢ TM-40M
CONCEPTS_GUIDE_TM40J_PROGRAM_MANAGER.md	[CNCPT]	↔ TM-40J [TM]
SYLLABUS_TM40J.md	[SYL]	→ TM-10, → TM-20, → TM-30 (required)
EX_40J_program_manager/EXERCISE.md	[EX]	→ TM-40J [TM]
EXAM_TM40J_PRE.md	[EXAM]	assesses TM-40J readiness
EXAM_TM40J_POST.md	[EXAM]	assesses TM-40J completion

Advanced track: → TM-50J **Inbound references from:** TM-40F (PM readiness/portfolio), TM-50J (prereq, recommends TM-40G/H/M/K/L), DATA_LITERACY_technical_reference, CHEATSHEET, task_index.html (18 tasks), mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ Deck_02_Scope_Engineering

TM-40K: Knowledge Manager

File	Type	Outbound Dependencies
TM_40K_KNOWLEDGE_MANAGER.md	[TM]	→ TM-10, → TM-20, → TM-30 (all required), ⇢ TM-40H (AIP Agent Studio), ⇢ TM-40L (TypeScript/OSDK, custom logic)
CONCEPTS_GUIDE_TM40K_KNOWLEDGE_MANAGER.md	[CNCPT]	↔ TM-40K [TM]
SYLLABUS_TM40K.md	[SYL]	→ TM-10, → TM-20, → TM-30 (required)
EX_40K_knowledge_manager/EXERCISE.md	[EX]	→ TM-40K [TM]
EXAM_TM40K_PRE.md	[EXAM]	assesses TM-40K readiness
EXAM_TM40K_POST.md	[EXAM]	assesses TM-40K completion

Advanced track: → TM-50K **Inbound references from:** TM-40F (KM info products), TM-50K (prereq), DATA_LITERACY_technical_reference, CHEATSHEET, task_index.html (6 tasks), mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ Controlled_Vocabularies · Identity_Who_Owns_The_Key · Deck_03_RDF_OWL_Foundations · Deck_05_Identity_Governance

TM-40L: Software Engineer

File	Type	Outbound Dependencies
TM_40L_SOFTWARE_ENGINEER.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40L; noted peer to TM-40G/H/M
CONCEPTS_GUIDE_TM40L_SOFTWARE_ENGINEER.md	[CNCPT]	role matrix shows TM-40L as SWE; ↔ TM-40L [TM]
SYLLABUS_TM40L.md	[SYL]	→ TM-10, → TM-20, → TM-30
EX_40L_software_engineer/EXERCISE.md	[EX]	→ TM-40L [TM]
EXAM_TM40L_PRE.md	[EXAM]	assesses TM-40L readiness
EXAM_TM40L_POST.md	[EXAM]	assesses TM-40L completion

Advanced track: → TM-50L **Inbound references from:** TM-40B (system admin/platform), TM-40E (pipeline construction), TM-40K (custom logic), TM-50L (prereq, TM-40H/M recommended), DATA_LITERACY_technical_reference, CHEATSHEET, task_index.html (30 tasks), mss_info_app/index.html, README

Recommended CDA decks [PPT]: ⇢ Deck_03_RDF_OWL_Foundations · Deck_04_Object_Type_Varieties · Deck_06_Relationship_Modeling · Deck_07_Temporal_Bitemporal · Data_Modeling_Fundamentals_Level2 · Deck_12_Capstone_Foundry

---

TM-40N: UI/UX Designer

File	Type	Outbound Dependencies
TM_40N_UX_DESIGNER.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40N; ↔ TM-40J (balanced team), ↔ TM-40L (design-to-dev handoff), ↔ TM-40O (platform constraints)
CONCEPTS_GUIDE_TM40N_UX_DESIGNER.md	[CNCPT]	role matrix shows TM-40N as Designer; ↔ TM-40N [TM]
SYLLABUS_TM40N.md	[SYL]	→ TM-10, → TM-20, → TM-30
EX_40N_ux_designer/EXERCISE.md	[EX]	→ TM-40N [TM]
EXAM_TM40N_PRE.md	[EXAM]	assesses TM-40N readiness
EXAM_TM40N_POST.md	[EXAM]	assesses TM-40N completion

Advanced track: → TM-50N **Cross-track companions:** ↔ TM-40J (PM — balanced team, backlog handoff), ↔ TM-40L (SWE — design implementation), ↔ TM-40O (Platform — performance constraints) **ASF alignment:** Maps to ASF UI/UX Designer (Product Designer) role. Covers Soldier Centered Design (SCD), user research, information architecture, visual design, prototyping, usability testing, accessibility (Section 508 / WCAG 2.1 AA), and design governance.

DRAFT

TM-40O: Platform Engineer

File	Type	Outbound Dependencies
TM_40O_PLATFORM_ENGINEER.md	[TM]	→ TM-10, → TM-20, → TM-30, → CONCEPTS_GUIDE_TM40O; ↔ TM-40L (platform consumers), ↔ TM-40H (AI compute infra), ↔ TM-40N (design constraints)
CONCEPTS_GUIDE_TM40O_PLATFORM_ENGINEER.md	[CNCPT]	role matrix shows TM-40O as Platform Eng; ↔ TM-40O [TM]
SYLLABUS_TM40O.md	[SYL]	→ TM-10, → TM-20, → TM-30
EX_40O_platform_engineer/EXERCISE.md	[EX]	→ TM-40O [TM]
EXAM_TM40O_PRE.md	[EXAM]	assesses TM-40O readiness
EXAM_TM40O_POST.md	[EXAM]	assesses TM-40O completion

Advanced track: → TM-50O **Cross-track companions:** ↔ TM-40L (SWE — primary platform consumer), ↔ TM-40H (AI Eng — model serving infrastructure), ↔ TM-50L Ch.6 (DevSecOps from application perspective — companion reading) **ASF alignment:** Maps to ASF Platform Engineer role. Covers platform-as-product, Kubernetes, IaC/GitOps, container security (Iron Bank), CI/CD pipeline design, deployment strategies, air-gapped/DDIL operations, and RMF/ATO from infrastructure perspective. References ASF CReATE platform architecture.

SECTION 4 — ADVANCED SPECIALIST TRACKS (TM-50G–O)

Prereq for each: corresponding TM-40G–O (required)

Track	Prereq	Key Cross-Refs	Advanced Track File
TM-50G (Adv ORSA)	TM-40G	↔ TM-40H, TM-40M, TM-40J	TM_50G_ORSA_ADVANCED.md + CG
TM-50H (Adv AI Eng)	TM-40H	↔ TM-50G, TM-50M, TM-50J, TM-50K, TM-50L; ⇢ TM-40M	TM_50H_AI_ENGINEER_ADVANCED.md + CG
TM-50M (Adv ML Eng)	TM-40M	↔ TM-50H (fine-tuning infra)	TM_50M_ML_ENGINEER_ADVANCED.md + CG

Track	Prereq	Key Cross-Refs	Advanced Track File
TM-50J (Adv PM)	TM-40J	⇢ TM-40G, TM-40H, TM-40M, TM-40K, TM-40L, TM-40N	TM_50J_PROGRAM_MANAGER_ADVANCED.md + CG
TM-50K (Adv KM)	TM-40K	↔ TM-50H (corpus design)	TM_50K_KNOWLEDGE_MANAGER_ADVANCED.md + CG
TM-50L (Adv SWE)	TM-40L	⇢ TM-40H, TM-40M; ↔ TM-50H (OSDK integration)	TM_50L_SOFTWARE_ENGINEER_ADVANCED.md + CG
TM-50N (Adv UI/UX)	TM-40N	↔ TM-50J (portfolio strategy), ↔ TM-50L (design system impl), ↔ TM-50O (perf budgets)	TM_50N_UX_DESIGNER_ADVANCED.md + CG
TM-50O (Adv Platform)	TM-40O	↔ TM-50L (platform-app boundary), ↔ TM-50J (platform roadmap), ↔ TM-50N (portal design)	TM_50O_PLATFORM_ENGINEER_ADVANCED.md + CG

Syllabi: SYLLABUS_TM50G–O present and published **Exams:** EXAM_TM50G–O PRE + POST present **Exercises:** EX_50G–O directories present (EX_50G_orsa/, EX_50H_ai_engineer/, EX_50M_ml_engineer/, EX_50J_program_manager/, EX_50K_knowledge_manager/, EX_50L_software_engineer/, EX_50N_ux_designer/, EX_50O_platform_engineer/ — each with EXERCISE.md + ENVIRONMENT_SETUP.md)

SECTION 4B — ARCHITECTURE PUBLICATIONS (PDF)

Standalone reference material. These are published PDF outputs of the CDA, EA, GDAP, MIM, and Ontology architecture documents. No prerequisites — not part of the TM course sequence. Source markdown lives in `doctrine/` subdirectories (cross-referenced in Section 0B).

CDA — Cognitive Data Architecture

PDF	Type	Description
ARCH_CDA_OVERVIEW.pdf	[SUPP]	CDA enterprise index and user taxonomy
ARCH_CDA_DOCTRINE_OVERVIEW.pdf	[SUPP]	Doctrine-driven development overview
ARCH_CDA_DOCTRINE_AGENT.pdf	[SUPP]	Doctrine agent architecture
ARCH_CDA_AVT25_ASSESSMENT.pdf	[SUPP]	AVT-25 assessment framework

PDF	Type	Description
ARCH_CDA_IDENTITY_VS_CLASSIFICATION.pdf	[SUPP]	Identity vs classification design patterns
ARCH_CDA_CONSTRAINTS_AND_DIRECTIVES.pdf	[SUPP]	12 operational constraints and directives
ARCH_CDA_CANON_ADP_CROSSWALK.pdf	[CANON]	ADP crosswalk for all specialist tracks
ARCH_CDA_CANON_CONDITIONS.pdf	[CANON]	Conditions, indicators, and thresholds
ARCH_CDA_CANON_ENGAGEMENT.pdf	[CANON]	Engagement canon
ARCH_CDA_CANON_INFORMATION.pdf	[CANON]	Information canon
ARCH_CDA_AGENTS_OVERVIEW.pdf	[SUPP]	Agents subsystem index and hierarchy diagram
ARCH_CDA_AGENTS_CORE_PRINCIPLES.pdf	[SUPP]	12 bedrock principles: Stability Stack, Four-Layer Architecture, anti-patterns
ARCH_CDA_AGENTS_ONTOLOGY_ENGINEER.pdf	[SUPP]	Ontology engineer agent specialization (RDF, OWL2, SHACL, SPARQL)
ARCH_CDA_AGENTS_ENTITY_RESOLUTION.pdf	[SUPP]	9-stage entity resolution pipeline
ARCH_CDA_AGENTS_INGESTION_INTEGRATION.pdf	[SUPP]	Five-Stage Ingestion Pattern, ELT, Dagster DAG

Enterprise Architecture

PDF	Type	Description
ARCH_EA_00_REFERENCE_CARD.pdf	[EA]	EA quick-reference card
ARCH_EA_01_FOUNDATION.pdf	[EA]	EA foundational concepts
ARCH_EA_02_SCHOOLS_OF_THOUGHT.pdf	[EA]	EA schools of thought
ARCH_EA_03_ARTIFACTS_AND_VIEWS.pdf	[EA]	EA artifacts and views
ARCH_EA_04_GOVERNANCE.pdf	[EA]	EA governance
ARCH_EA_05_MILITARY_APPLICATION.pdf	[EA]	EA military application

DRAFT

GDAP — Global Doctrine Alignment Platform

PDF	Type	Description
ARCH_GDAP_OVERVIEW.pdf	[SUPP]	Platform quickstart, LlamaIndex retrieval stack, API endpoints
ARCH_GDAP_VISION.pdf	[SUPP]	Full vision: 10 use-case domains, 20 pipeline steps, DVEE
ARCH_GDAP_PERSISTENCE_STRATEGY.pdf	[SUPP]	DuckDB canonical truth, LlamaIndex serving artifacts, versioning policy
ARCH_GDAP_ACCEPTANCE_TESTS.pdf	[SUPP]	P0/P1/P2 release gates across 9 sections
ARCH_GDAP_ADR_0001_LLAMAINDEX.pdf	[SUPP]	ADR-0001: DuckDB/LlamaIndex boundary rules

MIM — MIP Information Model

PDF	Type	Description
ARCH_MIM_OVERVIEW.pdf	[SUPP]	Toolchain: HTML/XSD parsers, Foundry backend, repo structure
ARCH_MIM_STANDARD.pdf	[SUPP]	MIM semantic model: namespaces, types, roles, design principles
ARCH_MIM_STATE.pdf	[SUPP]	Project snapshot: maturity, passing/failing tests, priorities
ARCH_MIM_ACADEMICS.pdf	[SUPP]	NATO interoperability, MIM-Foundry alignment, UML-OT mapping
ARCH_MIM_DECISION_RECORDS.pdf	[SUPP]	ADR structure: repo-level and per-package decisions
ARCH_MIM_FUTURE_CLASSES.pdf	[SUPP]	Planned packages: mim-studio, mim-ui, adapters, backends, mim-sdk
ARCH_MIM_ONTOLOGY_DOCS.pdf	[SUPP]	OSDK Maker Package TypeScript API

DRAFT

Ontology

PDF	Type	Description
ARCH_ONTOLOGY_DESIGN_PRINCIPLES.pdf	[SUPP]	DDD, DRY, Open/Closed, PECS, Composition; Foundry-specific

SECTION 5 — DOCTRINE & STANDARDS

File	Type	Referenced By
DATA_LITERACY_technical_reference.md	Doctrine/ADP-analog	TM-10, TM-20, TM-30, TM-40G–O, TM-50G–O (all recommended prereq read)
DATA_LITERACY_senior_leaders.md	Doctrine/senior leader brief	Standalone; referenced in README, QUICK_START
GLOSSARY_data_foundry.md	Reference	Standalone; referenced in README, cheatsheet
NAMING_AND_GOVERNANCE_STANDARDS.md	Standards	EX-20, EX-30 (required read); TM-40M Appendix A, TM-40L Appendix B
USAREUR_AF_DATA_ANALYTICS_STRATEGY.md	Strategy/CG-signed	All TMs (strategic context); COMMANDERS_GUIDE, POLICY_LETTER, DATA_LITERACY_senior_leaders
UNIFIED_DATA_TRANSITION_STRATEGY.md	Strategy/CUI	TM-30, TM-40G–O, TM-50G–O (ODT quarterly product cycle; resource allocation)
cheatsheet.md	Quick Reference	Standalone; links to all TMs A–O, 50G–O, base tracks

SECTION 6 — TRAINING MANAGEMENT

File	Covers	Key Cross-Refs
MTP_MSS.md	Master training plan; course overview table + TM-10 throughput risk item (Sec 7-4, G-3 action required)	All TMs + T3-F/T3-I; corrected durations: TM-20=5d, TM-30=5d, TM-40A–F=3d, TM-40G/H/M/L=5d, TM-40J/K=4d, T3-F=0.5d, T3-I=5d

File	Covers	Key Cross-Refs
POI_MSS.md	Program of instruction; tier/prereq structure	All TMs; prereq tree A–O; TM-40J/K now correctly in TM-30 branch
CAD_MSS.md	Course admin; access reqs by track	All TMs; TM-40J/K prereq = TM-30 required
TEO_MSS.md	T&EO evaluation standards; TM20-03 Row 5 now [CRITICAL] (date arithmetic)	TM-10 through TM-40M critical performance measures
ANNUAL_TRAINING_SCHEDULE.md	Schedule by track (2026)	All TMs; TM-40J/K prereq = TM-30
ANNUAL_TRAINING_CALENDAR_FY27.html	Interactive FY27 training calendar (HTML)	Visual companion to ANNUAL_TRAINING_SCHEDULE
FACULTY_DEVELOPMENT_PLAN.md	Instructor quals by track	All TMs; WFF (A–F) vs. Specialist (G–O) instructor profiles
COMPLETION_CERTIFICATE.md	Certificate templates	Generic; course title filled at completion
CURRICULUM_MAINTENANCE_SOP.md	Maintenance SOP + Platform Monitoring procedure (Sec 2A) + Semi-Annual Deep Review (Sec 5A)	Scope: TM-10 through TM-40O
ENROLLMENT_SOP.md	Enrollment process + prereq verification + Training Records Requirements section (minimum data standard, retention, query process)	All TMs; TM-40J/K prereq = TM-30 required
COMMANDERS_GUIDE_MSS_TRAINING.md	Commander/XO reference: who trains on what, timelines, CDR responsibilities, throughput risk	→ ENROLLMENT_SOP; → MTP; all TM levels
lesson_plans/TM10/TM10_LESSON_PLANS.md	Lesson plan: TM-10	TM-10 Ch 1–8, DATA_LITERACY_technical_reference Ch 1
lesson_plans/TM20_LESSON_PLAN_OUTLINES.md	Lesson plan: TM-20	TM-20 Ch 2–5, NAMING_AND_GOVERNANCE_STANDARDS
lesson_plans/TM30_LESSON_PLAN_OUTLINES.md	Lesson plan: TM-30	TM-30 Ch 2–9, Standards Ch 3–4

DRAFT

File	Covers	Key Cross-Refs
lesson_plans/TM40_SPECIALIST_LESSON_PLAN_OUTLINES.md	Lesson plans: TM-40G–O	All specialist tracks G–O; WFF A–F managed separately with WFF proponents
POLICY_LETTER.md	Training policy letter	All tracks; command direction and training requirements by echelon
AAR_TEMPLATE.md	After Action Review template	All training events; standard format for post-course review
lesson_plans/LP_TEMPLATE.md	Standard lesson plan template	Blank template; TLO, Key Delivery Notes, Student Activity, Assessment blocks
ENTERPRISE_V10_PLAN.md	ODT Enterprise v10 release plan (2026-03-12)	5 bounded contexts (Architecture/Pipelines/Products/Foundry/QA); M1–M5 milestone gates; agent assignments; feature checklist
ENTERPRISE_IMPLEMENTATION_PLAN.md	Enterprise implementation plan — CUI	6-phase/20-week hardening plan; 46 acceptance functions; 12 CG architectural constraints; current-state assessment
FOUNDRY_BOOTCAMP_SOP.md	Foundry Bootcamp SOP: quarterly event enrollment, execution, evaluation	→ FBC_GUIDE, → ENROLLMENT_SOP, → ANNUAL_TRAINING_SCHEDULE
INSTRUCTOR_OVERVIEW.md	Instructor onboarding guide: program structure, materials, execution	All TMs + T3-F/T3-I; → MTP, → POI, → CAD, → TEO, → ENROLLMENT_SOP, → lesson plans
DEMO_BRIEF_CG.md	CG decision brief: request TASKORD authorization for MSS MTT	→ TASKORD, → WHITE_PAPER; ↔ all training management docs
TASKORD_MSS_TRAINING_CELL.md	Draft TASKORD: MSS MTT activation (theater-wide)	→ MTP, → POI, → ENROLLMENT_SOP, → ANNUAL_TRAINING_SCHEDULE
WHITE_PAPER_MSS_TRAINING.md	Strategic white paper: case for MSS training across USAREUR-AF	→ all TMs (program overview); ↔ DEMO_BRIEF_CG
AR_350_1_ALIGNMENT_MAP.md	AR 350-1 alignment map: full MSS ecosystem mapped to 11 regulatory areas; SAT artifact coverage; gap analysis with mitigations	→ MTP, → POI, → CAD, → TEO, → FACULTY_DEVELOPMENT_PLAN, → ENROLLMENT_SOP, → CURRICULUM_MAINTENANCE_SOP; ↔ all 15 analytics apps

DRAFT

File	Covers	Key Cross-Refs
MSS_TRAINING_BRIEF.pptx	CG briefing slide deck (PowerPoint)	↔ DEMO_BRIEF_CG (companion visual)
lesson_plans/TM40_WFF_LESSON_PLAN_OUTLINES.md	Lesson plans: TM-40A–F WFF tracks	All WFF tracks A–F
lesson_plans/TM50_ADVANCED_LESSON_PLAN_OUTLINES.md	Lesson plans: TM-50G–O advanced tracks	All advanced tracks G–O
lesson_plans/T3F_LESSON_PLAN_OUTLINES.md	Lesson plans: T3-F MSC Force Multiplier	T3-F course; TM-10 lesson plans (teach-back source)
lesson_plans/T3I_LESSON_PLAN_OUTLINES.md	Lesson plans: T3-I Instructor Certification	T3-I course; all TM-10/TM-20/TM-30 materials (microteaching source)
UNIT_DATA_TRAINER_SOP.md	UDT governance: authorities, limitations, reporting, recertification, commander nomination template	→ T3-F (certification requirement); → C2DAO (reporting chain); → ENROLLMENT_SOP

SECTION 7 — HTML APPLICATIONS

File	Contents	TM Dependencies
mss_info_app/index.html	Training hub — 6 WFF cards (A–F, TM-30 chip) + 8 specialist cards (G–O, TM-30 chip) + 8 TM-50 cards	All TM-40A–O PDFs, all CGs, all exams, all syllabi
mss_info_app/training_schedule.html	Schedule view	Training schedule data
task_index.html	Task index by specialist track	TM-40G(17), TM-40H(1), TM-40M(21), TM-40J(18), TM-40K(6), TM-40L(30) = 93 total; TM-40N/O pending integration
mss_info_app/index_sharepoint.	SharePoint-hosted variant of MSS Training Hub	Same as index.html; adapted for SharePoint embedding

File	Contents	TM Dependencies
html		
mss_info_app/PILOT_AGENT_PROMPT.md	System prompt for MSS pilot AI agent	↔ mss_info_app/index.html (serves the training hub)
mss_info_app/DEPLOYMENT.md	Deployment guide: Cloudflare Pages + SharePoint + Foundry	→ index.html, → index_sharepoint.html
mss_info_app/Maven_Rollout_Plan.pptx	Rollout plan slide deck (PPTX)	↔ DEMO_BRIEF_CG (CG rollout strategy)
source_material/course_portal/	CDA slide library — 29 decks, 3 tracks, 397 PNG slides	⇢ all TM levels (see Section 8); source of all [PPT] refs

SECTION 8 — CDA SLIDE LIBRARY (REPOS/)

Source: `maven_training/source_material/course_portal/` **Format:** Slide decks (PNG images + PDF/PPTX downloads); manifest-driven **Status:** Not yet integrated into mss_info_app; planned as panel-slide-library **Role in curriculum:** Conceptual prereq layer — fills the "why and what" gap the TM series does not cover

All 29 decks listed below. Track assignment and TM alignment are authoritative for merge planning.

Track: Intro To Data (12 decks)

Deck ID	Title	Slides	TM Alignment	Notes
army_data_orientation_v1	Army Data Orientation	14	⇢ Data Literacy (SL) / all personnel	Direct parallel to doctrine panel content; Army-specific framing
architecture_primer	Architecture Primer	12	⇢ before TM-10	Orientation to the data stack before first MSS login
2026_Data_Stack_Deep_Dive	2026 Data Stack — Five Layers	10	⇢ TM-20	Explains the five-layer stack TM-20 teaches users to build in
L2_Ingestion_Integration_Deep_Dive	L2 Ingestion & Integration	10	⇢ TM-20	Pipeline concepts behind Pipeline Builder tasks

Deck ID	Title	Slides	TM Alignment	Notes
The_Semantic_Layer_Instructions	The Semantic Layer	11	⇢ TM-20 / TM-30	Why the Ontology layer exists; frames TM-20 Ontology Manager tasks
What_Is_An_Ontology	What Is an Ontology?	12	⇢ TM-20	Conceptual base for Object Types; gap in TM-20
Data_Modeling_Foundations	Data Modeling Foundations	11	⇢ TM-20 / TM-30	Normalization → strategy; underpins TM-30 architecture decisions
Data_Platforms_Cloud_Deep_Dive	Data Platforms & Cloud	10	⇢ TM-20 / TM-30	Where Foundry/MSS sits in the broader platform landscape
L5_Activation_Interface_Deep_Dive	L5 Activation & Interface	10	⇢ TM-20	Workshop conceptual grounding; explains why Workshop widgets exist
Dont_Filter_This_Isnt_Excel	Don't Filter — This Isn't Excel	11	⇢ TM-20	Critical mindset shift; Foundry vs spreadsheet mental model
decision_advantage_deep_dive	Decision Advantage	10	⇢ TM-40A–F (all WFF)	The "why" behind WFF data integration; all six WFF tracks
AI_ML_Beyond_The_Hype	AI/ML — Beyond the Hype	11	⇢ TM-40H / TM-40M	Conceptual framing required before AIP Logic and model training

Track: Data 101 (7 decks)

Deck ID	Title	Slides	TM Alignment	Notes
The_Four_Layers	The Four Layers	13	⇢ TM-20 / TM-30	Data layer model; directly maps to stack diagram in TM-20
Data_Modeling_Fundamentals_Level1	Data Modeling Fundamentals — Level 1	15	⇢ TM-20	Object type design patterns; prereq conceptual content for TM-20 tasks
ObjectType_WhatToWatchFor	Object Types — What to Watch For	13	⇢ TM-20 / TM-30	Anti-patterns; fills gap neither TM-20 nor TM-30 covers
Links_and_Relationships	Links & Relationships	11	⇢ TM-20	Link type conceptual base; maps to "Create Link Types" in TM-20
Controlled_Vocabularies	Controlled Vocabularies	11	⇢ TM-30 / TM-40K	Vocab standardization; TM-40K owns this operationally

DRAFT

Deck ID	Title	Slides	TM Alignment	Notes
Identity_Who_Owns_The_Key	Identity — Who Owns the Key?	11	⇢ TM-30 / TM-40K / TM-40L	Primary keys, master data, identity governance
Dont_Filter_This_Isnt_Excel	(see Intro To Data above)	—	⇢ TM-20	Listed in both tracks in source manifest

Track: Data 201 (10 decks)

Deck ID	Title	Slides	TM Alignment	Notes
Semantic_Modelling_Course_Intro	Semantic Modelling — Course Intro	10	⇢ TM-30 / TM-40L	Entry to formal semantic modeling; TM-30 architects and SWEs
Deck_01_Architecture_First_Principles	01 — Architecture First Principles	10	⇢ TM-30 / TM-40L	Core architecture decision principles; above TM-30 procedure level
Deck_02_Scope_Engineering	02 — Scope Engineering	10	⇢ TM-30 / TM-40J	How to scope a data problem; TM-30 governance + PM track
Deck_03_RDF_OWL_Foundations	03 — RDF & OWL Foundations	10	⇢ TM-40K / TM-40L	RDF triples, OWL classes; knowledge graph and semantic layer work
Deck_04_Object_Type_Varieties	04 — Object Type Varieties	10	⇢ TM-30 / TM-40L	Advanced object type design patterns
Deck_05_Identity_Governance	05 — Identity & Governance	10	⇢ TM-30 / TM-40K	Formal governance patterns for shared ontology
Deck_06_Relationship_Modeling	06 — Relationship Modeling	10	⇢ TM-30 / TM-40L	Advanced link type and relationship design
Deck_07_Temporal_Bitemporal	07 — Temporal & Bitemporal	10	⇢ TM-40G / TM-40M / TM-40L	Time-aware data modeling; ORSA time series + SWE pipeline patterns
Data_Modeling_Fundamentals_Level2	Data Modeling Fundamentals — Level 2	13	⇢ TM-40G / TM-40M / TM-40L	Advanced modeling; senior technical specialist level
Deck_12_Capstone_Foundry	12 — Capstone: Foundry	10	⇢ TM-40G / TM-40H / TM-40L	Bridges CDA theory to Foundry/MSS practice; single highest-value deck for specialists

SECTION 9 — EXTERNAL REFERENCE LIBRARY: PALANTIR DEVELOPERS

Source: Palantir Developers YouTube channel (@PalantirDevelopers) — official Palantir product deep-dive video library. **Integration method:** (1) Inline NOTE callouts embedded in TM files at the relevant chapter/task locations. (2) SELF_STUDY_ADDENDUM.md files in each TM directory for organized self-study guidance. No class time added — all content is reference or optional self-study only. **[SS]** = Self-Study Addendum file type (used in this section only)

Self-Study Addenda — File Registry

File	Type	Supports	Groups / Videos
tm/TM_30_advanced_builder/SELF_STUDY_ADDENDUM.md	[SS]	TM-30	5 groups: Workshop Scenarios, Pipeline Monitoring, Quiver Advanced, Contour Advanced, Platform Security
tm/TM_40A_intelligence/SELF_STUDY_ADDENDUM.md	[SS]	TM-40A	5 groups: Ontology & Data Foundations, Quiver Analysis, Geospatial/Mapping, AI-Assisted Intelligence & RAG, Platform Security & Access (19 videos)
tm/TM_40B_fires/SELF_STUDY_ADDENDUM.md	[SS]	TM-40B	5 groups: Ontology & Data Foundations, Quiver Analysis & Targeting, Contour Dashboards, Platform Architecture, AI & Automation (14 videos)
tm/TM_40C_movement_maneuver/SELF_STUDY_ADDENDUM.md	[SS]	TM-40C	5 groups: Ontology & Data Foundations, Quiver Analysis, Contour Dashboards, Platform Architecture & Real-Time, AI & Automation (14 videos)
tm/TM_40D_sustainment/SELF_STUDY_ADDENDUM.md	[SS]	TM-40D	5 groups: Ontology & Data Foundations, Quiver Analysis & LOGSTAT, Contour Dashboards, Supply Chain Patterns, Platform Architecture (15 videos)
tm/TM_40E_protection/SELF_STUDY_ADDENDUM.md	[SS]	TM-40E	5 groups: Ontology & Data Foundations, Quiver Analysis & Risk, Contour Dashboards, Platform Architecture & Security, AI-Assisted Operations (15 videos)
tm/TM_40F_mission_command/SELF_STUDY_ADDENDUM.md	[SS]	TM-40F	5 groups: Ontology & Data Foundations, Quiver Staff Analysis, Contour Dashboards, Platform Architecture & COP Design, AI & Decision Support (16 videos)
tm/TM_40G_orsa/SELF_STUDY_ADDENDUM.md	[SS]	TM-40G	4 groups: Quiver analytics, Contour analytics, Platform Architecture, AI/Advanced

File	Type	Supports	Groups / Videos
tm/TM_40H_ai_engineer/SELF_STUDY_ADDENDUM.md	[SS]	TM-40H	9 groups: RAG, Evals, MCP, Observability, Safety, Agent Studio, AIP Logic, Ontology, Platform
tm/TM_40M_ml_engineer/SELF_STUDY_ADDENDUM.md	[SS]	TM-40M	7 groups: Spark, PySpark testing, Iceberg, Compute Modules, Pipeline Monitoring, Model Registry, Feature Engineering
tm/TM_40J_program_manager/SELF_STUDY_ADDENDUM.md	[SS]	TM-40J	4 groups: Delivery & Scaling, Project Tracking, Case Studies, Leadership
tm/TM_40K_knowledge_manager/SELF_STUDY_ADDENDUM.md	[SS]	TM-40K	5 groups: RAG & Retrieval, Knowledge Architecture, Semantic Search, AIP-Assisted KM, Platform Admin
tm/TM_40L_software_engineer/SELF_STUDY_ADDENDUM.md	[SS]	TM-40L	8 groups: OSDK 2.0, Functions, Code Repos, Cipher, MCP, Edge Ontology, Spark, Platform SDK (~45 videos)
tm/TM_40N_ux_designer/SELF_STUDY_ADDENDUM.md	[SS]	TM-40N	5 groups: Workshop/application surfaces, Ontology/data model, platform capabilities, accessibility/security, case studies
tm/TM_40O_platform_engineer/SELF_STUDY_ADDENDUM.md	[SS]	TM-40O	5 groups: CI/CD pipeline ops, security/compliance, pipeline monitoring, platform architecture, case studies
tm/TM_50G_orsa_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50G	TM-40G groups + 7 TM-50G-specific additions (Bayesian, persistent ORSA, data layer)
tm/TM_50H_ai_engineer_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50H	TM-40H groups + 8 TM-50H-specific additions (multi-agent, observability, enterprise)
tm/TM_50M_ml_engineer_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50M	8 groups, 29 videos (Iceberg advanced, xAI model tuning, Lightweight Transforms, Scaling AI)
tm/TM_50J_program_manager_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50J	TM-40J groups + enterprise autonomy, process orchestration, enterprise automation
tm/TM_50K_knowledge_manager_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50K	TM-40K groups + Advanced Ontology, Enterprise KM, multimodal data plane

DRAFT

File	Type	Supports	Groups / Videos
tm/TM_50L_software_engineer_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50L	TM-40L reference + 5 TM-50L additions (Code-based AI, interoperability, Gallatin observability, Edge Embedded, Developer Experience)
tm/TM_50N_ux_designer_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50N	TM-40N reference + 5 TM-50N additions (design systems, DDIL/edge, AI interfaces, governance, strategic context)
tm/TM_50O_platform_engineer_advanced/SELF_STUDY_ADDENDUM.md	[SS]	TM-50O	TM-40O reference + 5 TM-50O additions (fleet mgmt, observability/reliability, RMF automation, DevEx, strategic leadership)

Inline NOTE Callouts — Summary by Track

Track	NOTE Count	Primary Chapters
TM-30	4	Ch 2 (Workshop Scenarios), Ch 3 (Pipeline Monitoring), Ch 5 (Quiver), Ch 7 (Security)
TM-40G	6	Ch 8 (Quiver/Contour analytics)
TM-40H	12	Ch 2, 5, 6, 7, 8, 9 (RAG, Evals, MCP, Observability, Safety)
TM-40M	10	Spark, PySpark testing, Iceberg, Compute Modules, Pipeline Monitoring
TM-40J	6	Phase 5/6 lifecycle, project tracking tools, delivery planning
TM-40K	9	Knowledge architecture, AIP/RAG, search/discovery
TM-40L	30	All chapters (OSDK 2.0, Functions, Code Repos, Cipher, MCP, Edge Ontology, Spark, Platform SDK)
TM-50G	3	Ch 4 Bayesian, Ch 8 persistent ORSA, data layer
TM-50H	5	Ch 2 multi-agent, Ch 6 observability, Ch 8 enterprise
TM-50M	4	Iceberg, xAI model tuning, Lightweight Transforms, Scaling AI
TM-50J	3	Enterprise autonomy, process orchestration, enterprise automation

DRAFT

Track	NOTE Count	Primary Chapters
TM-50K	3	Advanced Ontology, Enterprise KM, multimodal data plane
TM-50L	5	Code-based AI dev, interoperability, Gallatin observability, Edge Embedded, Developer Experience
Total	**~100**	Across TM-30 through TM-50M

CHANGE LOG — ISSUES FOUND & FIXED (12 MARCH 2026 AUDIT)

#	File	Line	Issue	Fix Applied
1	TM_40G_ORSA.md	81	"see TM-30 and TM-40G/40B" — TM-40B is WFF Fires, not pipeline dev	Changed to "see TM-30"
2	SYLLABUS_TM40J.md	9	TM-30 listed as "recommended but not required" — contradicts TM source	Changed to "TM-10/20/30 all required"
3	SYLLABUS_TM40K.md	9	TM-30 listed as "recommended" — contradicts TM source	Changed to "TM-10/20/30 all required"
4	POI_MSS.md	38, 51–52	TM-40J/K listed outside TM-30 branch; labeled TM-30 as exception	Fixed prereq tree; TM-40J/K now under TM-30 branch
5	CAD_MSS.md	71–72, 75	TM-40J/K prereq listed as "TM-30 recommended but not required"	Updated to "TM-30 required"; advisory note updated
6	ENROLLMENT_SOP.md	63–64	TM-40J/K prereq listed as TM-10/20 only	Updated to include TM-30 required
7	MTP_MSS.md	58–66	Multiple duration errors: TM-20=2d (should be 5), TM-30=3d (should be 5), TM-40G/H/M/L=4d (should be 5), TM-40J/K prereq=TM-20 (should be TM-30)	All durations and prereqs corrected
8	ANNUAL_TRAINING_SCHEDULE.md	120, 133, 188	TM-40J/K prereq listed as TM-20; prereq chain table omitted TM-30	Updated all three locations
9	(PDF) 42 stale PDFs	—	Old specialist naming scheme (e.g., A=ORSA instead of G=ORSA; stale 50-series A–F existed)	Deleted all 42 stale PDFs

#	File	Line	Issue	Fix Applied
10	build_pdfs.py	—	Missing WFF syllabi A–F, exams A–F, exercises A–F	Added all to MD_TARGETS and PUB_TYPES

AUDIT VERIFICATION COMMANDS

```
# Run full corpus audit (from repo root)
python3 scripts/audit.py

# Rebuild PDFs for changed source files
python3 scripts/build_pdfs.py

# Verify PDF count and check for stale files
ls maven_training/pdf/*.pdf | wc -l
```

Expected audit result: PASS — 0 issues

CHANGE LOG — ISSUES FOUND & FIXED (13 MARCH 2026 SECOND FULL AUDIT)

Scope: 925 files, 6 parallel agent passes. Full read of every file in corpus.

#	File(s)	Issue	Fix Applied
11	source_material/course_portal/assets/The Semantic Layer Instructions/	Space in directory name (naming violation)	Renamed → The_Semantic_Layer_Instructions/
12	source_material/cda_docs/training/Data 201/	Space in directory name	Renamed → removed (was empty after dedup)
13	source_material/cda_docs/training/Intro To Data/	Spaces in directory name	Renamed → removed (was empty after dedup)
14	source_material/course_portal/downloads/The Semantic Layer Instructions.pdf	Space in filename	Renamed → The_Semantic_Layer_Instructions.pdf

#	File(s)	Issue	Fix Applied
15	source_material/cda_docs/training/The Semantic Layer Instructions.pdf	Space in filename	Renamed → The_Semantic_Layer_Instructions.pdf
16	source_material/cda_docs/reference/Cross-Domain Solutions_ Complete Technical Reference for Ontology Modeling.pdf	Spaces in filename	Renamed → cross-domain-solutions-technical-reference-ontology-modeling.pdf
17	source_material/cda_docs/training/Intro To Data/The Semantic Layer Instructions.pdf	Identical duplicate of training/ root copy	Deleted
18	source_material/cda_docs/training/Data 201/Data_Modeling_Fundamentals_Level2.pdf	Identical duplicate of training/ root copy	Deleted
19	source_material/cda_docs/reference/DDOF_Playbook_v2_0.pdf	Identical duplicate of cda_docs/ root copy	Deleted
20	course_portal/manifest.json + downloads/index.html	Stale paths referencing renamed files	Updated to match new filenames
21	DEPENDENCY_MAP.md	Missing entries: CDA_CONSTRAINTS_AND_DIRECTIVES, POLICY_LETTER, AAR_TEMPLATE, LP_TEMPLATE	Added to Sections 0B and 6

OPEN ITEM — RESOLVED:

#	Files Affected	Issue	Resolution
OPEN-1	TM_40A–F [TM], CONCEPTS_GUIDE_TM40A–F [CG], SYLLABUS_TM40A–F [SYL], mss_info_app, mss_widget, DEPENDENCY_MAP	WFF tracks prereq inconsistency — some files stated TM-20 only, others TM-30.	**RESOLVED 2026-03-13:** Design decision — all WFF tracks (TM-40A–F) now require TM-30 (same prereq chain as Specialist tracks, TM-10 + TM-20 + TM-30). All governance docs and app files updated. TM/CG content files require separate update pass to align.

CHANGE LOG — RALF BRANCH MIGRATION (14 MARCH 2026)

Scope: 25 files migrated from `orsuh/repos` `ralf` branch (12 commits ahead of main, unmerged). Content classified as TM-40/50 specialist level (architecture doctrine, engineering reference).

#	Files Added	Destination	Level
22	doctrine/CG_GUIDANCE.md	Senior leader doctrine	TM-10/20
23	doctrine/cda_doctrine/CDA_OVERVIEW.md	CDA user taxonomy + enterprise index	TM-30 bridge
24	doctrine/cda_doctrine/agents/CDA_AGENTS_OVERVIEW.md	Agents index doc	TM-40H/M/K/L
25	doctrine/cda_doctrine/agents/CDA_AGENTS_CORE_PRINCIPLES.md	12 bedrock principles, 14 anti-patterns, decision framework	TM-40H/M/K/L
26	doctrine/cda_doctrine/agents/CDA_AGENTS_ONTOLOGY_ENGINEER.md	Specializes Principles 3/7/8/9/10/11; RDF/OWL2/SHACL/SPARQL	TM-40H/K/L
27	doctrine/cda_doctrine/agents/CDA_AGENTS_ENTITY_RESOLUTION.md	9-stage ER pipeline; threshold bands; data product architecture	TM-40H/M/L
28	doctrine/cda_doctrine/agents/CDA_AGENTS_INGESTION_INTEGRATION.md	Five-Stage Ingestion Pattern; Dagster DAG; VAULTIS-A	TM-40H/M/L
29	doctrine/ONTOLOGY_DESIGN_PRINCIPLES.md	DDD, DRY, Open/Closed, PECS, Composition; Foundry-specific rules	TM-40H/K/L
30	doctrine/gdap/GDAP_OVERVIEW.md	LlamaIndex retrieval stack, API endpoints	TM-40H/M/K/L
31	doctrine/gdap/GDAP_VISION.md	Full 10-domain vision, 20 pipeline steps, DVEE	TM-40H/M/K/L
32	doctrine/gdap/GDAP_PERSISTENCE_STRATEGY.md	DuckDB vs LlamaIndex boundary; versioning/rebuild policy	TM-40H/M/L
33	doctrine/gdap/GDAP_ACCEPTANCE_TESTS.md	P0/P1/P2 release gates across 9 sections	TM-40H/M/L

DRAFT

#	Files Added	Destination	Level
34	doctrine/gdap/GDAP_ADR_0001_LLAMAINDEX.md	ADR-0001: DuckDB = truth, LlamaIndex = serving	TM-40H/M/L
35	doctrine/mim/MIM_OVERVIEW.md	Toolchain: parsers, Foundry backend, package structure	TM-40H/M/K/L
36	doctrine/mim/MIM_STANDARD.md	MIM semantic model, namespaces, roles, code types, Semantic IDs	TM-40H/M/K/L
37	doctrine/mim/MIM_STATE.md	Project snapshot 2026-02-28; maturity by area; gaps/priorities	TM-40H/M/L
38	doctrine/mim/MIM_ACADEMICS.md	Dr. Gerz / NATO; MIM↔Foundry structural alignment analysis	TM-40H/M/K/L
39	doctrine/mim/MIM_DECISION_RECORDS.md	ADR: repo-level and per-package ADR structure	TM-40H/L
40	doctrine/mim/MIM_FUTURE_CLASSES.md	Planned adapters, backends, mim-studio, mim-sdk	TM-40H/M/L
41	doctrine/mim/MIM_ONTOLOGY_DOCS.md	OSDK Maker Package full TypeScript API reference	TM-40H/L
42	quick_reference/cda_reference/DOCTRINE_ELEMENT_FOUNDRY_MAPPING.md	Property mapping, bi-temporal fields, TS interface, OSDK queries, enums	TM-40H/M/L
43	training_management/ENTERPRISE_V10_PLAN.md	ODT v10 release plan; 5 bounded contexts; M1–M5 gates	TM-40H/M/L
44	training_management/ENTERPRISE_IMPLEMENTATION_PLAN.md	6-phase/20-wk hardening plan — **CUI**	TM-40H/M/L

Note on ENTERPRISE_IMPLEMENTATION_PLAN.md: Source document bears `CUI` classification. Handle per local policy before committing to shared repository.

CHANGE LOG — PALANTIR DEVELOPERS ENRICHMENT (14 MARCH 2026)

Scope: Full Palantir Developers YouTube channel (@PalantirDevelopers) enumerated via yt-dlp (~130 videos). All videos mapped to TM tracks (TM-30 through TM-50M). TM-30 scope limited to UI-only tools (no code). TM-40/50 specialist tracks received code-level content appropriate to their domains.

Method: (1) Inline NOTE callouts embedded at relevant chapter/task anchor points. (2) SELF_STUDY_ADDENDUM.md overflow files for content that did not fit inline. No class time or schedule changes made.

#	Files Added / Modified	Change	Level
45	TM_30_ADVANCED_BUILDER.md	4 inline NOTEs added (Workshop Scenarios, Pipeline Monitoring, Quiver advanced, Security)	TM-30
46	SYLLABUS_TM30.md	Supplemental self-study note added after Day 1 evening reading block	TM-30
47	tm/TM_30_advanced_builder/SELF_STUDY_ADDENDUM.md	NEW — 5 groups covering TM-30-scoped UI tool depth content (~15 videos)	TM-30
48	TM_40G_ORSA.md	6 inline NOTEs added (Quiver/Contour analytics)	TM-40G
49	tm/TM_40G_orsa/SELF_STUDY_ADDENDUM.md	NEW — 4 groups, 16 videos	TM-40G
50	TM_40H_AI_ENGINEER.md	12 inline NOTEs added (RAG, Evals, MCP, Observability, Safety)	TM-40H
51	tm/TM_40H_ai_engineer/SELF_STUDY_ADDENDUM.md	NEW — 9 groups, 43 videos	TM-40H
52	TM_40M_ML_ENGINEER.md	10 inline NOTEs added (Spark, PySpark, Iceberg, Compute Modules)	TM-40M
53	tm/TM_40M_ml_engineer/SELF_STUDY_ADDENDUM.md	NEW — 7 groups, 25 videos	TM-40M
54	TM_40J_PROGRAM_MANAGER.md	6 inline NOTEs added (lifecycle phases, project tracking, delivery)	TM-40J
55	tm/TM_40J_program_manager/SELF_STUDY_ADDENDUM.md	NEW — 4 groups	TM-40J
56	TM_40K_KNOWLEDGE_MANAGER.md	9 inline NOTEs added (knowledge architecture, AIP/RAG, search)	TM-40K

#	Files Added / Modified	Change	Level
57	tm/TM_40K_knowledge_manager/SELF_STUDY_ADDENDUM.md	NEW — 5 groups	TM-40K
58	TM_40L_SOFTWARE_ENGINEER.md	30 inline NOTEs added across all chapters (OSDK, Functions, Cipher, MCP, Edge, Spark)	TM-40L
59	tm/TM_40L_software_engineer/SELF_STUDY_ADDENDUM.md	NEW — 8 groups, ~45 videos	TM-40L
60	TM_50G_ORSA_ADVANCED.md	3 inline NOTEs added	TM-50G
61	tm/TM_50G_orsa_advanced/SELF_STUDY_ADDENDUM.md	NEW — extends TM-40G + 7 TM-50G additions	TM-50G
62	TM_50H_AI_ENGINEER_ADVANCED.md	5 inline NOTEs added	TM-50H
63	tm/TM_50H_ai_engineer_advanced/SELF_STUDY_ADDENDUM.md	NEW — extends TM-40H + 8 TM-50H additions	TM-50H
64	TM_50M_ML_ENGINEER_ADVANCED.md	4 inline NOTEs added	TM-50M
65	tm/TM_50M_ml_engineer_advanced/SELF_STUDY_ADDENDUM.md	NEW — 8 groups, 29 videos	TM-50M
66	TM_50J_PROGRAM_MANAGER_ADVANCED.md	3 inline NOTEs added	TM-50J
67	tm/TM_50J_program_manager_advanced/SELF_STUDY_ADDENDUM.md	NEW — extends TM-40J + enterprise content	TM-50J
68	TM_50K_KNOWLEDGE_MANAGER_ADVANCED.md	3 inline NOTEs added	TM-50K
69	tm/TM_50K_knowledge_manager_advanced/SELF_STUDY_ADDENDUM.md	NEW — extends TM-40K + advanced	TM-50K
70	TM_50L_SOFTWARE_ENGINEER_ADVANCED.md	5 inline NOTEs added	TM-50L
71	tm/TM_50L_software_engineer_advanced/SELF_STUDY_ADDENDUM.md	NEW — TM-50L additions + reference to TM-40L addendum	TM-50L

DRAFT

#	Files Added / Modified	Change	Level
72	scripts/build_pdfs.py	Added PUB_TYPES + MD_TARGETS for: TM-50 syllabi (6), EX_50 exercises (6), SELF_STUDY_ADDENDUM files (13)	Build
73	DEPENDENCY_MAP.md	Section 7 added for Palantir Developers external reference library	All

CHANGE LOG — RALF BRANCH FULL INVENTORY (14 MARCH 2026)

Scope: Full file inventory of ralf-branch-migrated content revealed additional files beyond the initial 25 tracked in entries #22–44. These files were already present in Section 0B but not captured in the changelog.

#	Files Added	Destination	Level
74	doctrine/CDA_CONSTRAINTS_AND_DIRECTIVES.md	12 CDA operational constraints; governs all specialist CDA work	TM-40G–O
75	doctrine/cda_doctrine/CDA_DOCTRINE_OVERVIEW.md	Doctrine-driven development framework; DDD applied to CDA	TM-40G–O
76	doctrine/cda_doctrine/CDA_DOCTRINE_AGENT.md	Doctrine Agent reference — AI-assisted doctrine authoring patterns	TM-40H, TM-40L
77	doctrine/cda_doctrine/CDA_AVT25_ASSESSMENT.md	AVT-25 assessment framework for CDA capability evaluation	TM-40G, TM-40H
78	doctrine/cda_doctrine/CDA_IDENTITY_VS_CLASSIFICATION.md	Identity vs. classification disambiguation; semantic precision	TM-30, TM-40K, TM-40L
79	doctrine/enterprise_architecture/EA_00_REFERENCE_CARD.md	EA one-pager; frameworks overview (TOGAF, NAF, DODAF, Zachman)	TM-30, TM-40K, TM-40L
80	doctrine/enterprise_architecture/EA_01_FOUNDATION.md	EA foundational concepts; what EA is and is not	TM-30
81	doctrine/enterprise_architecture/EA_02_SCHOOLS_OF_THOUGHT.md	TOGAF, Zachman, FEA, NAF, DODAF compared	TM-30, TM-40K
82	doctrine/enterprise_architecture/EA_03_ARTIFACTS_AND_VIEWS.md	NAF/DODAF views; architecture artifacts; diagram types	TM-40K, TM-40L
83	doctrine/enterprise_architecture/EA_04_GOVERNANCE.md	Architecture governance; review boards; decision authority	TM-40K

#	Files Added	Destination	Level
84	doctrine/enterprise_architecture/EA_05_MILITARY_APPLICATION.md	EA applied to military C2 / joint operations context	TM-30, TM-40G–O
85	doctrine/cda_doctrine/canon/CANON_ADP_CROSSWALK.md	ADP 3-0 / 6-0 crosswalk to Foundry data model	TM-40G–O
86	doctrine/cda_doctrine/canon/CANON_CONDITIONS_INDICATORS_THRESHOLDS.md	Conditions, indicators, and threshold patterns for data products	TM-40G, TM-40H
87	doctrine/cda_doctrine/canon/CANON_ENGAGEMENT.md	Engagement authority and engagement event data model	TM-40G, TM-40H
88	doctrine/cda_doctrine/canon/CANON_INFORMATION.md	Information taxonomy and schema for common operational data	TM-40H, TM-40K
89	quick_reference/cda_reference/EA_VS_DA.md	Enterprise Architecture vs. Data Architecture decision tool	TM-30, TM-40K
90	quick_reference/cda_reference/ENTERPRISE_DATA_COMPASS.md	Data strategy decision compass; orientation for PM/KM roles	TM-40J, TM-40K
91	quick_reference/cda_reference/LESSONS_LEARNED.md	CDA implementation lessons learned registry	All tracks
92	quick_reference/cda_reference/PLAN_FOR_SUCCESS.md	CDA project planning checklist and success criteria	TM-40J, TM-30

CHANGE LOG — PDF BUILD: ARCH / MGMT / REF SERIES (14 MARCH 2026)

Scope: 42 new PDFs generated for all ralf-branch architecture, management, and reference docs. Distinct naming convention (`ARCH_`, `MGMT_`, `REF_`) to clearly separate from TM/CG/SYL/EX/EXAM corpus.

PDF naming scheme:

Prefix	Category	PUB_TYPE label	Example stems
`ARCH_CDA_`	CDA Doctrine & Agents	ARCHITECTURE REFERENCE / ODT-CDA	ARCH_CDA_AGENTS_CORE_PRINCIPLES, ARCH_CDA_CANON_ADP_CROSSWALK
`ARCH_GDAP_`	GDAP Platform	ARCHITECTURE REFERENCE / ODT-GDAP	ARCH_GDAP_OVERVIEW, ARCH_GDAP_ADR_0001_LLAMAINDEX

Prefix	Category	PUB_TYPE label	Example stems
`ARCH_MIM_`	MIM Reference	ARCHITECTURE REFERENCE / ODT-MIM	ARCH_MIM_STANDARD, ARCH_MIM_ACADEMICS
`ARCH_EA_`	Enterprise Architecture	ARCHITECTURE REFERENCE / ODT-EA	ARCH_EA_01_FOUNDATION, ARCH_EA_05_MILITARY_APPLICATION
`ARCH_ONTOLOGY_`	Ontology Design	ARCHITECTURE REFERENCE / ODT-ONT	ARCH_ONTOLOGY_DESIGN_PRINCIPLES
`MGMT_`	Management / Planning	SENIOR LEADER GUIDANCE or ENTERPRISE PLAN	MGMT_CG_GUIDANCE, MGMT_ENTERPRISE_V10_PLAN
`REF_`	Quick Reference	QUICK REFERENCE / ODT-REF	REF_DOCTRINE_ELEMENT_FOUNDRY_MAPPING, REF_EA_VS_DA

#	Action	Files	Output
93	scripts/build_pdfs.py — PUB_TYPES + MD_TARGETS expanded	Added ARCH_CDA/GDAP/MIM/EA/ONT, MGMT_, REF_ key prefixes and 42 source paths	42 new PDFs in maven_training/pdf/
94	DEPENDENCY_MAP.md	Updated header metadata; added changelog entries #74–94	This file

CHANGE LOG — PRE-V2.0 FULL-CORPUS AUDIT (14 MARCH 2026)

Scope: Team-of-teams content accuracy and consistency audit across all curriculum files, React widget, HTML app, Python scripts, and PPTX presentation files. Primary goal: verify prereq chain enforcement (TM-30 hard prereq for ALL TM-40 tracks A–F and G–O) and content alignment between all delivery surfaces.

Audit findings and resolutions:

#	File	Issue	Resolution
95	README.md (root)	WFF prereq stated TM-20 in 4 places (table, tree, footnote lines)	Fixed → TM-30 in all 4 locations
96	maven_training/training_management/POLICY_LETTER.md	Official governance table: WFF tracks listed TM-20 as prereq	Fixed → TM-30

#	File	Issue	Resolution
97	maven_training/mss_widget/src/panels/Home.tsx	WFF specialist row "Start Here" cell showed TM-10 → TM-20 only; TM-30 missing	Fixed → TM-10 → TM-20 → TM-30 chain
98	maven_training/quick_reference/cheatsheet.md	WFF track description said "after TM-20"; prereq chain diagram wrong	Fixed → TM-30 throughout; diagram rebuilt with WFF + Specialist both branching from TM-30
99	maven_training/mss_widget/src/panels/TM40.tsx	TM-40J card: builder-focused language ("PM Dashboards, Reporting Pipelines") mismatched actual TM content (Agile PM theory, ML lifecycle, stakeholder mgmt); BLUF embedded full WFF track list creating confusion	Fixed: TM-40J card rewritten to PM theory; BLUF simplified with cross-reference to HOME tab
100	maven_training/mss_info_app/index.html	TM-40J card matched stale builder-focused TM40.tsx (dual-maintenance surface)	Fixed: TM-40J card updated to match TM40.tsx (PM theory content)
101	.gitignore	node_modules/ missing; 65MB React deps at risk of commit	Added node_modules/
102	maven_training/pdf/MSS_Project_Overview.pptx	Slide 1: old "6 Specialist Tracks: 40A ORSA · 40B AI · 40C MLE..." scheme	Rebuilt via python-pptx: "12 Tracks (WFF + Specialist) / WFF (A–F): Intel · Fires · M&M / Specialist (G–O): ORSA · AI Eng · MLE · PM · KM · SWE · UX · PE"
103	maven_training/pdf/MSS_Training_Progression.pptx	Specialist column: original 6-slot design insufficient; partial fix left WFF A–D mixed with specialist H–I, sub-headers misplaced	Full column rebuild via python-pptx: deleted all 19 old content shapes; clean layout with "▸ WFF TRACKS (A–F)" (A–F) / "▸ SPECIALIST (G–O)" (G–O) sub-headers; TM-50 G–O note added at bottom
104	scripts/build_pdfs.py (PDF rebuild)	3 source files changed: POLICY_LETTER.md, cheatsheet.md, task_index.html	Hash-based rebuild: 3 PDFs regenerated (POLICY_LETTER.pdf, CHEATSHEET.pdf + task_index.html-derived); 199 unchanged
105	DEPENDENCY_MAP.md	Audit findings not logged	This entry

DRAFT

#	File	Issue	Resolution
106	SYLLABUS_TM40B.md, TM40_WFF_LESSON_PLAN_OUTLINES.md, DEPENDENCY_MAP.md	ATP 3-60 (inactive, May 2015) referenced instead of superseding FM 3-60 (Aug 2023)	replace_all ATP 3-60 → FM 3-60 in all 3 files (2 + 4 + 1 = 7 instances)
107	DEPENDENCY_MAP.md	No external doctrinal reference index	Added "External Doctrinal References" section with 32-pub table
108	TM_40E, SYLLABUS_TM40E, DEPENDENCY_MAP, index.html, index_sharepoint.html, TM40_WFF_LESSON_PLAN_OUTLINES	Phantom FM 3-37 cited (pub does not exist); protection capstone is ADP 3-37	Removed/replaced FM 3-37 → ADP 3-37 across 6 files (10 edits); zero instances remain
109	TM_50K_KNOWLEDGE_MANAGER_ADVANCED	FM 6-01.1 cited (pub type wrong); correct is ATP 6-01.1	Changed FM 6-01.1 → ATP 6-01.1 (Techniques for Effective Knowledge Management)
110	TM_40C, CONCEPTS_GUIDE_TM40C, index.html, index_sharepoint.html	FM 3-90-1 and FM 3-90-2 cited; both merged into FM 3-90 (May 2023)	Consolidated 19 refs across 4 files → FM 3-90, Offense and Defense (May 2023)
111	TM_40C_MOVEMENT_MANEUVER	Stale edition dates: FM 3-0 (2022), FM 3-09 (2020), FM 5-0 (2022), FM 6-0 (2014)	Updated to current: FM 3-0 (Mar 2025), FM 3-09 (Aug 2024), FM 5-0 (Nov 2024), FM 6-0 (May 2022)
112	TM_40K, TM_50K	FM 7-0 subtitle "Train to Win in a Complex World" (2016 ed.)	Updated to current: FM 7-0, Training (Jun 2021)
113	TM_50H_AI_ENGINEER_ADVANCED	AR 25-55 cited (rescinded; incorporated into AR 25-30)	Changed AR 25-55 → AR 25-30 (FOIA)
114	25 TM/doctrine/training_management files	Strategic guidance docs mixed into Governing/Doctrinal References tables	Separated into distinct "Strategic Guidance" sections; 21 files modified

DRAFT

#	File	Issue	Resolution
115	13 TM files (TM-10, TM-40A–O)	No professional reading lists	Added Professional Reading List appendices with 65+ curated journal articles
116	7 files (CG docs, TM-40F/G/H/M, CDA)	No EUCOM theater context	Added Theater and Strategic Context sections (BRAVO Hackathon, Thunderforge, Posture Statement, 49B career path)
117	TM-40A, TM-40C, TM-40K, TM-40L, TM-50K, TM-50L, GLOSSARY	NATO MIM standards gap — no STANAG 5643/5527, ADatP-36/5644, or CWIX references	Added 4 NATO standards to doctrine refs, CWIX context notes, glossary MIM disambiguation
118	doctrine/, DEPENDENCY_MAP, index.html, Doctrine.tsx	USAREUR-AF Data and Analytics Strategy (CG-signed, May 2025) not in corpus	Created USAREUR_AF_DATA_ANALYTICS_STRATEGY.md; added to Sec 0B (Command Strategy), Sec 5, External Refs, doctrine panel (HTML + React), publications index
119	doctrine/, DEPENDENCY_MAP, index.html, Doctrine.tsx	Unified Data Transition Strategy quarterly product cycle (CUI) not in corpus	Created UNIFIED_DATA_TRANSITION_STRATEGY.md; added to Sec 0B, Sec 5, External Refs, doctrine panel (HTML + React), publications index

CHANGE LOG — STRUCTURAL ADDITIONS (18 MARCH 2026)

Scope: New course tracks (T3-I, T3-F, TM-40N/O, TM-50N/O), ASF role alignment, and WFF self-study addenda.

#	Files Added / Modified	Change	Level
120	tm/T3_F_msc_force_multiplier/ (TM, CG), syllabi/SYLLABUS_T3F.md, exercises/EX_T3F_force_multiplier/ (EXERCISE, ENV_SETUP), exams/EXAM_T3F_PRE/POST.md, lesson_plans/T3F_LESSON_PLAN_OUTLINES.md, training_management/UNIT_DATA_TRAINER_SOP.md, PDFs (8)	T3-F MSC Force Multiplier (Unit Data Trainer Certification). Half-day course, prereq TM-20. Authorizes TM-10 delivery, TM-10 exam proctoring.	T3-F

#	Files Added / Modified	Change	Level
121	tm/T3_I_instructor_certification/ (TM, CG), syllabi/SYLLABUS_T3I.md, exercises/EX_T3I_instructor_certification/ (EXERCISE, ENV_SETUP), exams/EXAM_T3I_PRE/POST.md, lesson_plans/T3I_LESSON_PLAN_OUTLINES.md, PDFs (8)	NEW — T3-I Instructor Certification (C2DAO Instructor Pipeline). Phase 1: 5-day classroom, prereq TM-30 + OIC selection. Phase 2: supervised practicum.	T3-I
122	tm/TM_40N_ux_designer/ (TM, CG, SS), syllabi/SYLLABUS_TM40N.md, exercises/EX_40N_ux_designer/ (EXERCISE, ENV_SETUP), exams/EXAM_TM40N_PRE/POST.md, PDFs (~8)	NEW — TM-40N UI/UX Designer specialist track. ASF Designer role alignment. Covers SCD, user research, IA, visual design, prototyping, usability testing, Section 508/WCAG 2.1 AA.	TM-40N
123	tm/TM_40O_platform_engineer/ (TM, CG, SS), syllabi/SYLLABUS_TM40O.md, exercises/EX_40O_platform_engineer/ (EXERCISE, ENV_SETUP), exams/EXAM_TM40O_PRE/POST.md, PDFs (~8)	NEW — TM-40O Platform Engineer specialist track. ASF Platform Engineer role alignment. Covers K8s, IaC/GitOps, Iron Bank, CI/CD, DDIL ops, RMF/ATO.	TM-40O
124	tm/TM_50N_ux_designer_advanced/ (TM, CG, SS), tm/TM_50O_platform_engineer_advanced/ (TM, CG, SS), syllabi/SYLLABUS_TM50N.md, SYLLABUS_TM50O.md, exercises/EX_50N_ux_designer/, EX_50O_platform_engineer/, exams/EXAM_TM50N/O_PRE/POST.md, PDFs (~12)	NEW — TM-50N (Advanced UI/UX) and TM-50O (Advanced Platform Eng) tracks. Prereqs: TM-40N, TM-40O respectively.	TM-50N/O
125	tm/TM_40A–F (6 SELF_STUDY_ADDENDUM.md files)	NEW — WFF self-study addenda for all six WFF tracks (Intelligence, Fires, Movement & Maneuver, Sustainment, Protection, Mission Command). 5 groups each, 14–19 videos per track. Palantir Developers source.	TM-40A–F

CHANGE LOG — ATIS REGISTRATION PACKET (16 MARCH 2026)

Scope: Build TR 350-70 compliant ATIS course registration packet for USAREUR-AF G3/7 submission (Phase 1) with forward compatibility for T2COM institutional registration (Phase 2).

#	File	Issue / Action	Resolution
118	scripts/build_atis_packet.py	New script: extracts all 21 course records from POI/CAD/MTP/TEO data into structured ATIS format	Outputs JSON + TR 350-70 formatted markdown; covers all fields: hours by method, blocks, T&EO crosswalk, equipment, prereqs, GO/NO-GO criteria
119	maven_training/atis/atis_courses.json	New file: machine-readable structured course data (63KB)	21 courses, 664 total program hours, full prereq tree
120	maven_training/atis/ATIS_COURSE_PACKET.md	New file: TR 350-70 formatted ATIS registration packet (1,477 lines)	Per-course records with all required ATIS fields; T2COM readiness notes embedded
121	maven_training/pdf/ATIS_COURSE_PACKET.pdf	New PDF: Army doctrine-styled ATIS packet	Generated via build_pdfs.py pipeline; cover page, DRAFT header/footer, page numbers
122	scripts/build_pdfs.py	ATIS packet not in PDF build pipeline	Added PUB_TYPES entry (ATIS_COURSE) + MD_TARGETS entry for atis/ATIS_COURSE_PACKET.md
123	DEPENDENCY_MAP.md	ATIS files not tracked	Added atis/ entries to Section 6; this changelog

CHANGE LOG — DDOF PLAYBOOK V2.2 RECONCILIATION (17 MARCH 2026)

Scope: Reconcile corpus with authoritative DDOF Playbook v2.2 (December 2025, T2COM C2DAO / HQDA CIO/G-6 / SAIS-ADD, CUI // FEDCON). Key changes: VAULTIS-A dimension names corrected, DDOF acronym expansion corrected, phase timelines updated, version references updated from v2.0 to v2.2.

VAULTIS-A authoritative definition (v2.2, p.9 glossary p.16): Visible, Accessible, Understandable, Linked, Trusted, Interoperable, Secure, Auditable. Supersession: VAUTI (5, AR 25-1 2019) → VAULTIS (7, DoD Data Strategy 2020) → VAULTIS-A (8, DDOF Playbook v2.2 2025).

Phase timelines (v2.2 Quick Reference Guide, p.14): Phase 1: 3-5 days | Phase 2: 5-10 days | Phase 3: 7-12 days | Phase 4: 15-25 days | Phase 5: 7-12 days | Total IOC: 30-60 days.

New v2.2 content not yet integrated into TM series (noted for future integration): - Genesis Mission Alignment (3 directives: Decision Dominance, Bureaucracy Elimination, Accountability) - Secretary of the Army Priorities mapping (Warfighter First, Speed Over Perfection, Cut the Fat, Trust the Troops) - ADC Registration Requirements for Phase 6 - Configuration Management Friction Matrix - Data Product Retirement criteria (90 days no access → FDM review, 180 days → retirement, quality <70% → remediate/retire) - Roles table (Decision Maker O-6+, C2DAO O-4/O-5, FDM O-3/O-4, etc.) - Platform-agnostic note: "works with Vantage, ADVANA, or any approved system"

#	File	Issue / Action	Resolution
124	source_material/cda_docs/ddof/reference/ddof.ts	VAULTIS-A property names wrong (Valid/Accurate/Unique/Timely/Integrated)	Renamed to Visible/Accessible/Understandable/Trusted/Interoperable per v2.2
125	doctrine/cda_doctrine/agents/CDA_AGENTS_CORE_PRINCIPLES.md	Principle 12 VAULTIS-A dimensions wrong	Updated 8 dimensions to match v2.2 authoritative definition
126	doctrine/cda_doctrine/agents/CDA_AGENTS_INGESTION_INTEGRATION.md	VAULTIS-A table dimensions wrong	Updated 8 dimensions to match v2.2
127	quick_reference/cda_reference/ENTERPRISE_DATA_COMPASS.md	VAULTIS-A dimensions wrong; phase timelines wrong; DDOF v2.0 ref	Fixed all three; v2.0 → v2.2 with correct authority line
128	quick_reference/cda_reference/PLAN_FOR_SUCCESS.md	DDOF v2.0 references	Updated to v2.2 (2 occurrences)
129	source_material/cda_apps/enterprise-data-compass/index.html	VAULTIS-A dimensions wrong; phase timelines wrong; DDOF v2.0	Fixed all three
130	source_material/cda_apps/plan-for-success/data/plan.json	DDOF v2.0 reference	Updated to v2.2
131	source_material/cda_apps/plan-for-success/data/radar.json	DDOF v2.0 reference	Updated to v2.2
132	doctrine/GLOSSARY_data_foundry.md	No VAULTIS-A entry; VAUTI entries did not note supersession	Added VAULTIS-A definition; updated VAUTI/VAUTI Framework entries with supersession chain; added VAULTIS/VAULTIS-A abbreviations

#	File	Issue / Action	Resolution
133	tm/TM_30_advanced_builder/TM_30_ADVANCED_BUILDER.md	DDOF = "Doctrine-Driven Ontology Framework" (wrong)	Corrected to "Defense Data Orchestration Framework"
134	source_material/README.md	DDOF = "Data-Driven Operational Framework" (wrong); v2.0 refs	Corrected acronym; updated to v2.2 with metadata (Dec 2025, T2COM C2DAO, CUI // FEDCON)
135	tm/_archive_pre_review/TM_30_ADVANCED_BUILDER.md	DDOF = "Doctrine-Driven Ontology Framework" (wrong)	Corrected to "Defense Data Orchestration Framework"

CHANGE LOG — DOCTRINE SOURCE RECONCILIATION (17 MARCH 2026, SESSION 2)

Scope: Comprehensive audit of corpus against authoritative source documents (43 ADPs/FMs/strategic docs downloaded and read). VAUTI→VAULTIS-A corrections across all active TMs, doctrine docs, and training management. DDOF lifecycle content added. Army doctrine (ADP 6-0, ADP 3-13) cognitive hierarchy integrated. WFF doctrine references validated and gaps filled.

#	File	Issue	Resolution
136	tm/TM_10_MAVEN_USER.md	VAUTI refs (3); no VAULTIS-A consumer section; no cognitive hierarchy	Updated governance chain to VAULTIS/VAULTIS-A; added Section 5-5 VAULTIS-A consumer literacy; added Section 1-2a cognitive hierarchy (ADP 6-0/3-13)
137	doctrine/DATA_LITERACY_technical_reference.md	Section 4-8 used VAUTI (5 dims); no VAULTIS-A; no DDOF Playbook ref; no cognitive hierarchy	Rewrote Section 4-8 as VAULTIS-A (8 dims, full targets); added DDOF Playbook to strategic refs; added Section 1-2 cognitive hierarchy; updated glossary entries
138	doctrine/DATA_LITERACY_senior_leaders.md	Section 1-4 used VAUTI (5 dims); glossary/refs stale	Rewrote Section 1-4 as VAULTIS-A (8 dims); updated glossary and references with supersession chain; added DDOF Playbook ref
139	tm/TM_20_BUILDER.md	VAUTI refs (2)	Updated AR 25-1 and DoD Data Strategy refs with VAULTIS-A supersession
140	tm/TM_30_ADVANCED_BUILDER.md	VAUTI refs (2); no DDOF lifecycle section	Updated refs; added Section 1-10 DDOF Lifecycle (6 phases, timelines, Genesis Mission, quality gates, retirement triggers)

#	File	Issue	Resolution
141	tm/TM_40K_KNOWLEDGE_MANAGER.md	VAUTI ref (1)	Updated to VAULTIS-A with DDOF Playbook citation
142	tm/TM_40L_SOFTWARE_ENGINEER.md	VAUTI refs (2)	Updated governance ref and glossary to VAULTIS-A
143	tm/TM_40M_ML_ENGINEER.md	VAUTI ref (1)	Updated to VAULTIS-A with DDOF Playbook citation
144	tm/TM_40A–F (all 6 WFF TMs)	No DDOF/VAULTIS-A references anywhere	Added DDOF Playbook v2.2 row to Strategic Guidance tables in all 6 files
145	tm/TM_40A_INTELLIGENCE.md	FM 3-84 (IPOE) not cited	Added FM 3-84 to Appendix H doctrine cross-reference
146	tm/TM_40E_PROTECTION.md	FM 3-37 not cited as standalone	Added FM 3-37 alongside ADP 3-37 in Table 1-4
147	tm/TM_50J_PROGRAM_MANAGER_ADVANCED.md	DDOF retirement triggers not cited	Added DDOF Playbook v2.2 automatic retirement trigger table (90/180-day, <70% quality)
148	training_management/COMMANDERS_GUIDE_MSS_TRAINING.md	Missing DoD Data Strategy, DDOF, UDRA, Army Data Plan details	Added 4 strategic guidance rows: DoD Data Strategy (VAULTIS), DDOF Playbook v2.2, UDRA v1.1, expanded Army Data Plan with SO7/SE05 workforce mandate
149	DEPENDENCY_MAP.md	DoD Data Strategy row said "VAUTI Framework"	Updated to "VAULTIS Framework (7 dimensions)"; added DDOF Playbook v2.2 row
150	tm/CONCEPTS_GUIDE_TM20_BUILDER.md	VAUTI ref (1)	Updated to VAULTIS/VAULTIS-A

SECTION 9 — ANALYTICS & TOOLING APPS (`APPS/`)

Source: `apps/` (repo root) **Stack:** Python, SQLAlchemy (SQLite/WAL), FastAPI (API), Streamlit (dashboards) **Portal:** `apps/portal.py` — unified landing page connecting all 15 apps (port 8500)

All `.db` files are gitignored. Seed data is generated via `python3 -m apps.<name>.seed`. Each app follows a standard structure: `db.py`, `models.py`, `api.py`, `dashboard.py`, `seed.py`.

Training Analytics (core 3)

App	API Port	Dashboard Port	Purpose	TM Dependencies
readiness_tracker	8001	8501	Unit/Soldier completion across prereq chain; RAG heatmap; bottleneck analysis; training funnel	→ All TMs (prereq chain is authoritative); ↔ enrollment_manager, ↔ instructor_manager
exam_analytics	8002	8502	Pre/post exam gain scores; item discrimination; question difficulty	→ All exams (TM-10 through TM-50); ↔ readiness_tracker
aar_aggregator	8003	8503	AAR trend analysis by WFF; frequency × severity matrix; recurring issues	→ All WFF tracks (TM-40A–F); ↔ lessons_learned

Training Operations

App	API Port	Dashboard Port	Purpose	TM Dependencies
progress_tracker	8004	8504	Individual training timelines; stalled Soldier detection; overdue milestones	→ readiness_tracker (prereq chain); ↔ enrollment_manager
mtt_scheduler	8005	8505	MTT event scheduling across AOR; instructor/venue allocation; conflict detection	↔ instructor_manager, ↔ enrollment_manager
data_quality	8010	8510	Pipeline health monitoring; quality metrics (5 types); anomaly alerting	↔ all pipeline-producing apps

Content & Quality Tools

App	API Port	Dashboard Port	Purpose	TM Dependencies
xref_validator	8006	8506	Cross-reference validation; broken link detection; prereq chain audit	→ all maven_training/ files
glossary_search	8007	8507	Full-text search across glossary terms and doctrine definitions	→ GLOSSARY_data_foundry.md, all doctrine/
offline_packager	—	8508	Bundle TMs/exercises/PDFs for DDIL environments	→ all maven_training/ content
sharepoint_sync	—	8509	Sync state tracking between local content and Cloudflare/SharePoint	→ all maven_training/ files

Faculty & Enrollment

App	API Port	Dashboard Port	Purpose	TM Dependencies
instructor_manager	8011	8511	Instructor certification tracking; course coverage matrix; teaching workload; expiration alerts	→ FACULTY_DEVELOPMENT_PLAN, → INSTRUCTOR_OVERVIEW; ↔ mtt_scheduler, ↔ enrollment_manager
enrollment_manager	8012	8512	Class enrollment; seat allocation; waitlists; roster management	→ ENROLLMENT_SOP; ↔ readiness_tracker, ↔ mtt_scheduler, ↔ instructor_manager

Knowledge Management

App	API Port	Dashboard Port	Purpose	TM Dependencies
curriculum_tracker	8013	8513	Document version tracking; review cycles; content freshness; SHA-256 change detection	→ CURRICULUM_MAINTENANCE_SOP; → all maven_training/ .md files
lessons_learned	8014	8514	Structured lessons-learned pipeline; TTP/WFF/echelon tagging taxonomy; action tracking	→ TM-40K (taxonomy design); ↔ aar_aggregator

DRAFT

Executive

App	API Port	Dashboard Port	Purpose	TM Dependencies
training_metrics	8015	8515	CG/DCG/G6 aggregation dashboard; risk register; BLUF briefing export; snapshot comparison	→ all 14 other apps (read-only aggregation); → TM-50J (senior leader reporting)

App Dependency Graph

```
training_metrics (CG Dashboard)
  ├── readiness_tracker ←── enrollment_manager
  ├── exam_analytics            ↑
  ├── aar_aggregator ←─── lessons_learned
  ├── progress_tracker
  ├── mtt_scheduler ←──── instructor_manager
  ├── data_quality
  ├── instructor_manager
  ├── enrollment_manager
  ├── curriculum_tracker
  └── lessons_learned
```

CHANGE LOG — ANALYTICS APPS EXPANSION (17 MARCH 2026)

Scope: Added 5 new analytics apps to close operational gaps identified in gap analysis against FACULTY_DEVELOPMENT_PLAN, ENROLLMENT_SOP, CURRICULUM_MAINTENANCE_SOP, TM-40K (taxonomy), and TM-50J (senior leader reporting). Portal updated from v3.0 (10 apps) to v4.0 (15 apps).

#	File(s)	Issue / Action	Resolution
151	apps/instructor_manager/ (7 files)	No tool to track instructor certifications, course coverage gaps, or teaching workload	New app: 3 tables (Instructor, Certification, TeachingHistory), coverage matrix, expiration tracking, workload analysis
152	apps/enrollment_manager/ (7 files)	No tool to manage class seats, waitlists, or rosters — MTT Scheduler handles events but not enrollment	New app: 3 tables (TrainingClass, Enrollment, WaitlistEntry), auto-waitlist, priority promotion, fill-rate tracking
15	apps/curriculum_tracker/ (7	No tool to track document freshness, review cycles, or detect content	New app: 3 tables (Document, ReviewCycle, ChangeLog), SHA-256 scanning of

#	File(s)	Issue / Action	Resolution
3	files)	changes	maven_training/, stale content detection
154	apps/lessons_learned/ (7 files)	AAR Aggregator limited to WFF-level rollup; TM-40K taxonomy (TTP/echelon/doctrine ref) not implemented	New app: 4 tables (Lesson, LessonTag, ActionItem, LessonComment), full tagging taxonomy, co-occurrence matrix, action tracking
155	apps/training_metrics/ (7 files)	No executive dashboard for CG/DCG/G6; no snapshot comparison or risk aggregation	New app: aggregates from all 14 apps, auto-generated risk register, BLUF briefing export, weekly snapshots
156	apps/portal.py	Portal only covered 10 apps	Updated: 3 new sections (Faculty & Enrollment, Knowledge Management, Executive), second KPI row, v3.0 → v4.0
157	DEPENDENCY_MAP.md	No apps/ section	Added Section 9 — Analytics & Tooling Apps (15 apps, port map, dependency graph)

CHANGE LOG — DOCTRINE INTEGRATION — PHASE 2 (17 MARCH 2026, SESSION 3)

The following changes integrate extracted doctrine from ADPs, FMs, DDOF Playbook v2.2, Army Data Plan, and UDRA v1.1 into the training corpus. Research agents extracted content from 6 ADPs (3-0, 6-0, 3-13, 2-0, 3-37, 7-0), 14 FMs (2-0, 3-0, 3-09, 3-12, 3-34, 3-57, 3-60, 3-84, 3-96, 5-0, 6-0, 6-22, 1-02.1, 3-94), DDOF Playbook v2.2, Army Data Plan (2022), and UDRA v1.1.

#	File	Change	Source
151	TM_20_BUILDER.md	VAUTI → VAULTIS-A supersession (2 locations)	DoD Data Strategy / DDOF Playbook v2.2
152	TM_30_ADVANCED_BUILDER.md	VAUTI → VAULTIS-A supersession (2 locations)	DoD Data Strategy / DDOF Playbook v2.2
153	TM_40K_KNOWLEDGE_MANAGER.md	VAUTI → VAULTIS-A (8 dims, 85% gate)	DDOF Playbook v2.2

#	File	Change	Source
154	TM_40L_SOFTWARE_ENGINEER.md	VAUTI → VAULTIS-A (strategic guidance + glossary)	DDOF Playbook v2.2
155	TM_40M_ML_ENGINEER.md	VAUTI → VAULTIS-A	DDOF Playbook v2.2
156	CONCEPTS_GUIDE_TM20_BUILDER.md	"DoD VAUTI" → "DoD VAULTIS / VAULTIS-A"	DoD Data Strategy
157	DATA_LITERACY_senior_leaders.md	"Trustworthy" → "Trusted" (line 72)	DDOF Playbook v2.2
158	TM_40A–F (6 files)	DDOF Playbook v2.2 / VAULTIS-A added to Ch 1 strategic guidance + end-of-file refs	DDOF Playbook v2.2
159	TM_30_ADVANCED_BUILDER.md	Sections 1-10a–d: DDOF roles, SMART criteria, fail-closed, ADC registration	DDOF Playbook v2.2
160	DATA_LITERACY_technical_reference.md	Section 4-9: UDRA 7 quality dimensions	UDRA v1.1, Table 8
161	DATA_LITERACY_technical_reference.md	Chapter 11: Data mesh concepts (products, domains, computational governance, 6 services)	UDRA v1.1
162	DATA_LITERACY_technical_reference.md	Chapter 12: Army Data Plan SO1–SO11, SE01–SE05	Army Data Plan (2022)
163	DATA_LITERACY_technical_reference.md	6 new glossary entries	UDRA v1.1 / Army Data Plan
164	TM_40A_INTELLIGENCE.md	FM 2-0 data literacy NOTE, intel process→pipeline table, 7 quality standards	FM 2-0 (Oct 2023)
165	TM_40B_FIRES.md	D3A lifecycle, FIVE-O taxonomy, TTLODAC schema, CARVER scoring	FM 3-60 / FM 3-09

DRAFT

#	File	Change	Source
166	TM_40C_MOVEMENT_MANEUVER.md	PMESII-PT/METT-TC(I), force ratios, geospatial data, recon as data collection	ADP 3-0 / FM 5-0 / FM 3-98
167	TM_40C Appendix K	Restructured from narrative list to doctrine mapping table (16 pubs)	Multiple FMs
168	TM_40D_SUSTAINMENT.md	8 sustainment principles, 10 classes of supply framework, running estimate NOTE	ADP 4-0 / FM 4-0 / FM 6-0
169	TM_40E_PROTECTION.md	CVP analysis, OPSEC 5-step mapping, GMAD NOTE, doctrine table updated	ADP 3-37 / FM 3-13.3 / FM 3-34
170	TM_40F_MISSION_COMMAND.md	FM 6-0 IM 6-task mapping, info relevance→VAULTIS-A crosswalk, App D enhanced	FM 6-0 / FM 5-0
171	TM_10_MAVEN_USER.md	GMAD framework, Operations Process, METT-TC(I) NOTE	FM 3-34 / ADP 5-0 / ADP 3-0
172	COMMANDERS_GUIDE_MSS_TRAINING.md	ADP 7-0: 9 training principles, T/P/U mapping, 3 training domains	ADP 7-0 (Apr 2024)
173	scripts/build_pdfs.py	Title page centering (flex spacer after topbar)	N/A
174	TM_40G_ORSA.md	Assessment framework (MOE/MOP/indicators), CARVER scoring model, force ratio/combat power	FM 5-0 / FM 3-60
175	TM_40H_AI_ENGINEER.md	ADP 3-13 AI/ML doctrine NOTE, PED→pipeline mapping, UDRA VAULTIS-A governance NOTE	ADP 3-13 / FM 2-0 / UDRA v1.1
176	TM_40J_PROGRAM_MANAGER.md	DDOF friction matrix, roles/PM oversight table, portfolio health metrics, 4 glossary entries	DDOF Playbook v2.2
177	TM_40K_KNOWLEDGE_MANAGER.md	FM 6-0 KM 5-step process, 3 tasks, 6 tool categories; FM 6-22 domains; FM 3-57 CKI NOTE	FM 6-0 / FM 6-22 / FM 3-57

DRAFT

#	File	Change	Source
178	TM_30_ADVANCED_BUILDER.md	Section 1-10e: DDIL operations (Denied/Degraded/Intermittent/Limited)	Army Data Plan SO6/SO10 / UDRA v1.1 / ADP 6-0
179	DEPENDENCY_MAP.md	Changelog entries #151–179 for Session 3 doctrine integration	N/A
180	TM_40M_ML_ENGINEER.md	Sections 1-8/9-2a: DDOF Phases 4–5 ML lifecycle, VAULTIS-A model compliance, ADP 3-13 human-machine teaming WARNING	DDOF Playbook v2.2 / ADP 3-13 / UDRA v1.1
181	TM_40L_SOFTWARE_ENGINEER.md	Section 1-5b: Army Data Plan SO 7 DevSecOps→DDOF phase mapping; Section 1-7: UDRA 6-service architecture; metadata fields WARNING	Army Data Plan / UDRA v1.1
182	TM_50G_ORSA_ADVANCED.md	Section 5-9: multi-variable sensitivity analysis, Bayesian assessment updating, longitudinal trend analysis, OAWG NOTE	FM 5-0 / FM 3-60
183	TM_50H_AI_ENGINEER_ADVANCED.md	Section 5-12: multi-source data fusion, adversarial ML defense, model drift monitoring, DDIL deployment NOTE	ADP 3-37 / FM 2-0 / Army Data Plan
184	TM_50J_PROGRAM_MANAGER_ADVANCED.md	Section 9-6: multi-product dependency mapping, cascading friction analysis, resource optimization, DDOF retirement NOTE	DDOF Playbook v2.2
185	TM_50K_KNOWLEDGE_MANAGER_ADVANCED.md	Enterprise KM at theater scale, FM 6-0 five-step process cross-reference, federated ontology governance	FM 6-0 / UDRA v1.1
186	TM_50L_SOFTWARE_ENGINEER_ADVANCED.md	Computational governance as code, DDIL-resilient pipeline design, zero trust architecture	Army Data Plan / UDRA v1.1 / DoD ZTRA
187	TM_50M_ML_ENGINEER_ADVANCED.md	Model governance at portfolio scale, federated ML for coalition, DDOF Phase 6 model versioning	DDOF Playbook v2.2 / UDRA v1.1
188	CONCEPTS_GUIDE_TM50G–M (6 files)	NOTE callouts referencing new TM-40 doctrine content in each parent TM	N/A

#	File	Change	Source
189	CONCEPTS_GUIDE_TM40B_FIRES.md	NOTE referencing D3A, FIVE-O, TTLODAC, CARVER content in TM-40B	N/A
190	CONCEPTS_GUIDE_TM40C_MOVEMENT_MANEUVER.md	NOTE referencing PMESII-PT/METT-TC(I), force ratios, recon content in TM-40C	N/A
191	CONCEPTS_GUIDE_TM40D_SUSTAINMENT.md	NOTE referencing 8 principles, 10 supply classes, running estimate in TM-40D	N/A
192	CONCEPTS_GUIDE_TM40E_PROTECTION.md	NOTE referencing CVP, OPSEC 5-step, GMAD content in TM-40E	N/A
193	CONCEPTS_GUIDE_TM40F_MISSION_COMMAND.md	NOTE referencing FM 6-0 IM tasks, VAULTIS-A crosswalk, App D in TM-40F	N/A
194	TM40_SPECIALIST_LESSON_PLAN_OUTLINES.md	Doctrine content NOTEs for new CARVER, DDOF friction matrix material	FM 3-60 / DDOF Playbook v2.2
195	TM50_ADVANCED_LESSON_PLAN_OUTLINES.md	Doctrine content NOTEs for Bayesian, adversarial ML, cascading friction material	Multiple sources
196	EXAM_TM30_POST.md	6 new DDOF Playbook questions (Q16–Q21): VAULTIS-A dims, retirement triggers, phase order, SMART, fail-closed	DDOF Playbook v2.2
197	EXAM_TM40M_POST.md	5 new DDOF/VAULTIS-A/ML questions (Q16–Q20): MVP mandate, Linked dimension, Phase 4 gate, Auditable, ADP 3-13	DDOF Playbook v2.2 / UDRA v1.1 / ADP 3-13
198	TM40_WFF_LESSON_PLAN_OUTLINES.md	Doctrine content NOTEs for all 6 WFF tracks (A–F) Block 1	Multiple ADPs/FMs
199	SELF_STUDY_ADDENDUM (TM-30)	Group 5: DDOF and Operational Design Patterns (Sections 1-10a–e)	DDOF Playbook v2.2
200	SELF_STUDY_ADDENDUM (TM-40G/H/J/K)	New Doctrine References sections for 4 specialist self-study addenda	Multiple sources

DRAFT

#	File	Change	Source
201	SELF_STUDY_ADDENDUM (TM-50G/H/J/K)	New Doctrine References sections for 4 advanced self-study addenda	Multiple sources
202	TM_40H_AI_ENGINEER.md	Section 1-5a: Strategic Guidance subsection (ADP 3-13, DDOF, VAULTIS-A, UDRA, Army Data Plan, DoD Responsible AI)	Multiple strategic sources
203	CONCEPTS_GUIDE_TM40H_AI_ENGINEER.md	Updated NOTE to reference new Section 1-5a Strategic Guidance	N/A
204	EXAM_TM40A_POST.md	2 doctrine MC questions: intel process→pipeline mapping (Q9), 7 quality standards (Q10)	FM 2-0 / TM-40A
205	EXAM_TM40B_POST.md	2 doctrine MC questions: D3A lifecycle Deliver phase (Q9), CARVER Accessibility factor (Q10)	FM 3-60 / TM-40B
206	EXAM_TM40C_POST.md	2 doctrine MC questions: METT-TC(I) Informational addition (Q9), 3:1 force ratio (Q10)	ADP 3-0 / FM 5-0 / TM-40C
207	EXAM_TM40D_POST.md	2 doctrine MC questions: Anticipation principle→data (Q9), Class IX update freq (Q10)	ADP 4-0 / FM 4-0 / TM-40D
208	EXAM_TM40E_POST.md	2 doctrine MC questions: CVP 3 factors (Q9), OPSEC 5-step sequence (Q10)	ADP 3-37 / FM 3-13.3 / TM-40E
209	EXAM_TM40F_POST.md	2 doctrine MC questions: FM 6-0 IM 6 tasks (Q9), Timely→VAULTIS-A (Q10)	FM 6-0 / TM-40F
210	EXAM_TM40H_POST.md	2 doctrine MC questions: ADP 3-13 human-machine teaming (Q16), PED Exploitation→AI/ML (Q17)	ADP 3-13 / FM 2-0 / TM-40H
211	EXAM_TM40K_POST.md	2 doctrine MC questions: FM 6-0 KM 5-step Design (Q16), FM 3-57 CKI validation (Q17)	FM 6-0 / FM 3-57 / TM-40K
212	EXAM_TM40L_POST.md	2 doctrine MC questions: SO 7 DevSecOps→DDOF 30-day MVP (Q16), UDRA security-critical metadata (Q17)	Army Data Plan / UDRA v1.1 / TM-40L

DRAFT

#	File	Change	Source
213	DEPENDENCY_MAP.md	Changelog entries #202–213 for Session 4 exam + strategic guidance gap closure	N/A

EXTERNAL DOCTRINAL REFERENCES

Publications prefixed TR/TP are published by TRADOC at adminpubs.tradoc.army.mil, not DA APD (armypubs.army.mil).

Publication	Title	Referenced By
AR 350-1	Army Training and Leader Development	All training management docs
AR 350-10	Management of Army Individual Training Requirements and Resources	ATIS_COURSE_PACKET, ENROLLMENT_SOP
TR 350-70	Army Learning Policy and Systems (TRADOC)	POI, CAD, CURRICULUM_MAINTENANCE_SOP, INSTRUCTOR_OVERVIEW
TP 350-70-14	Training Development in Institutional Domain (TRADOC)	POI, CAD, CURRICULUM_MAINTENANCE_SOP
TP 350-70-7	Army Educational Processes (TRADOC)	POI, INSTRUCTOR_OVERVIEW
TP 350-70-3	Faculty and Staff Development Program (TRADOC)	INSTRUCTOR_OVERVIEW
DA PAM 25-40	Army Publishing: Action Officers' Guide	Publication standards (all docs)
AR 25-50	Preparing and Managing Correspondence	Army writing style (all docs)
AR 25-1	Army Information Technology	TM-40K, TM-40L, POLICY_LETTER, data governance
DA PAM 25-1-1	Army IT Implementation Instructions	TM-40K, TM-40L

DRAFT

Publication	Title	Referenced By
AR 25-400-2	Army Records Management Program	TM-40K
DA PAM 25-403	Army Guide to Recordkeeping	TM-40K
ATP 6-01.1	Techniques for Effective Knowledge Management	TM-40K, TM-50K
DA PAM 600-3	Officer Professional Development	TM-40G (FA 49 ORSA)
AR 5-11	Management of Army M&S	TM-40G, TM-50G
DA PAM 5-11	VV&A of Army Models and Simulations	TM-40G, TM-50G
ATP 5-0.3	Multi-Service TTP for Operation Assessment	TM-40G, TM-50G
AR 71-9	Warfighting Capabilities Determination	TM-40G
Army DIR 2024-03	Digital Engineering Policy	TM-40H, TM-40M, TM-40L, TM-50H, TM-50M, TM-50L
FM 3-12	Cyberspace Operations and EW	TM-40E, TM-40H, TM-40M, TM-40L
DA PAM 25-2-5	Software Assurance	TM-40H, TM-40M, TM-40L
FM 3-27	Army Global Integrated Fires	TM-40B
ATP 3-01.81	Counter-UAS Techniques	TM-40B
FM 3-81	Maneuver Enhancement Brigade	TM-40C
ATP 4-33	Maintenance Operations	TM-40D
FM 1-0	Human Resources Support	TM-40D
AR 525-2	The Army Protection Program	TM-40E
AR 530-1	Operations Security	TM-40E
ATP 5-0.1	Army Design Methodology	TM-40F
TC 6-0.2	Training the C2 WFF for BN/BDE/BCT	TM-40F
ATP 2-33.4	Intelligence Analysis	TM-40A
ATP 2-22.9-1	PAI Research and OSINT	TM-40A
FM 3-60	Army Targeting (Aug 2023)	TM-40A, TM-40B

DRAFT

Publication	Title	Referenced By
FM 2-0	Intelligence (Jul 2018)	TM-40A, SYLLABUS_TM40A
FM 3-0	Operations (Mar 2025)	TM-40C, SYLLABUS_TM40C
FM 3-09	Field Artillery Operations and Fire Support (Aug 2024)	TM-40B, SYLLABUS_TM40B
FM 3-90	Offense and Defense (May 2023)	TM-40C (consolidates former FM 3-90-1 / FM 3-90-2)
FM 3-96	Brigade Combat Team	TM-40C
FM 4-0	Sustainment	TM-40D, SYLLABUS_TM40D
FM 5-0	Army Planning and Orders Production (Nov 2024)	TM-40C, TM-40F
FM 6-0	Commander and Staff Organization and Operations (May 2022)	TM-40F, SYLLABUS_TM40F
FM 7-0	Training (Jun 2021)	TM-40K, TM-50K
ADP 3-0	Operations	TM-40C, SYLLABUS_TM40C, CANON_ADP_CROSSWALK
ADP 3-19	Fires	TM-40B
ADP 3-37	Protection	TM-40E, SYLLABUS_TM40E
ADP 4-0	Sustainment	TM-40D, SYLLABUS_TM40D
ADP 6-0	Mission Command	TM-40F, SYLLABUS_TM40F, CANON_ADP_CROSSWALK
AR 25-30	The Army Freedom of Information Act Program	TM-50H (consolidates former AR 25-55)
ATP 2-01	Planning Requirements and Assessing Collection	TM-40A, SYLLABUS_TM40A
ATP 2-01.3	Intelligence Preparation of the Battlefield	TM-40A
ATP 3-01.8	Combined Arms for Air Defense	TM-40B
ATP 3-09.42	Fire Support for the Brigade Combat Team	TM-40B, SYLLABUS_TM40B
ATP 3-09.50	Battalion-Level Fire Support	TM-40B
ATP 3-37.2	Antiterrorism	TM-40E, SYLLABUS_TM40E

DRAFT

Publication	Title	Referenced By
ATP 3-52.2	Airspace Control	TM-40B
ATP 3-90.90	Army Tactical Standard Operating Procedures	TM-40C, SYLLABUS_TM40C
ATP 4-0.1	Army Theater Distribution	TM-40D, SYLLABUS_TM40D
DoD/Joint Strategic Guidance		
DoD Data Strategy (2020)	Data as Strategic Asset / VAULTIS Framework (7 dimensions)	TM-10, TM-20, TM-30, doctrine/
DDOF Playbook v2.2 (Dec 2025)	VAULTIS-A (8 dimensions), 6-phase lifecycle, 85% quality gate	All TMs, doctrine/, training_management/
DoD Data, Analytics & AI Adoption Strategy (Nov 2023)	AI Hierarchy of Needs; adopt-buy-create	COMMANDERS_GUIDE, POLICY_LETTER, doctrine/
DoD Responsible AI Strategy (Jun 2024)	Five AI Ethical Principles (RETRG)	TM-40H, TM-40M, TM-50H, TM-50M
DoD Zero Trust Reference Architecture v2.0 (Jul 2022)	152 ZT activities across 7 pillars	TM-30
DoD AI Cybersecurity Risk Mgmt Guide	Secure AI/ML development	TM-40H, TM-40M, TM-50H, TM-50M
DoDI 5000.87	Software Acquisition Pathway	TM-40L, TM-50L, TM-40J, TM-50J
DoDD 3000.09	Autonomy in Weapon Systems (Jan 2023)	TM-40A–F
DoD Software Modernization Strategy (Feb 2022)	DevSecOps frameworks	TM-40L, TM-50L
JADC2 Strategy Summary (Mar 2022)	Cross-domain data integration	TM-40A–F, TM-40G, TM-50G
JCOIE	Operating in the Information Environment	TM-40F
USAREUR-AF Theater Guidance		
USAREUR-AF Data and Analytics Strategy (May 2025)	CG-signed; 4 strategic outcomes, VAULTIS, Cognitive Hierarchy, decision dominance	All TMs, COMMANDERS_GUIDE, POLICY_LETTER, doctrine/

DRAFT

Publication	Title	Referenced By
Unified Data Transition Strategy (CUI)	Quarterly product cycle: Problem ID → Bootcamp → CADs → PSB → PoC → Exercise	TM-30, TM-40G–O, TM-50G–O, training_management/
Army Strategic Guidance		
Army Data Plan (2022)	11 strategic objectives for data transformation	TM-10, TM-20, TM-30, COMMANDERS_GUIDE, POLICY_LETTER
Army Cloud Plan (2022)	ZT, secure dev, data-driven decisions	TM-10, TM-20, TM-30
UDRA v1.1 (Feb 2025)	Data mesh, decentralized governance	TM-30, TM-40G–O, TM-50G–O
Army CIO Data Stewardship Memo (Apr 2024)	Chain of responsibility for data governance	TM-10, TM-20, TM-30, TM-40K
Army Directive 2024-02	Agile Software Dev & Acquisition (Dec 2024)	TM-40L, TM-50L, TM-40J, TM-50J
NATO		
NATO Data Strategy for the Alliance (Feb 2025)	Alliance-wide data governance mandate	TM-30, TM-40K, TM-50K, CDA_CONSTRAINTS
NATO Data Centric Reference Architecture v2 (2025)	Digital transformation reference arch	TM-30, CDA_CONSTRAINTS
NATO Data Quality Framework (Aug 2025)	Quality governance and metrics	TM-30
NATO Digital Transformation Implementation Strategy (Oct 2024)	MDO interoperability	TM-40A–F
NATO Warfighting Capstone Concept (2021)	6 Critical Enablers incl. Data	TM-40F
ADatP-34/NISP (NATO)	C3 Interoperability Standards	TM-40K, TM-50K, TM-40L, TM-50L
STANAG 5636/NCMS (NATO)	Core Metadata Specification	TM-40K, TM-50K
STANAG 5643 (proposed)	MIM Governance Standard	TM-40K, TM-50K, TM-40L, TM-50L
ADatP-5644	Web Service Messaging Profile (WSMP)	TM-40L, TM-50L

DRAFT

Publication	Title	Referenced By
ADatP-36	Friendly Force Information (FFI)	TM-40A, TM-40C
STANAG 5527	Friendly Force Tracking Systems Interoperability	TM-40A

MSS TRAINING ANALYTICS SUITE

Toggle: Check the "Analytics Apps" checkbox in the HTML dependency map to overlay the app architecture on the corpus graph. **Standalone reference:** See `analytics_apps/DEPENDENCY_MAP.md` for full app catalog, ports, startup sequence, and cross-app dependency graph.

The analytics suite (15 Streamlit micro-apps + portal + shared theme) is integrated into the HTML dependency map as a toggleable layer on the far right. Apps are positioned by tier:

```
[HUB]         portal.py (8500)
[AGGREGATOR]  training_metrics (8015/8515)
[CORE]        readiness_tracker (8001/8501)
[DEPENDENT]   progress_tracker (8004/8504) → readiness_tracker
[STANDALONE]  exam_analytics, aar_aggregator, mtt_scheduler,
              xref_validator, glossary_search, offline_packager,
              sharepoint_sync, data_quality, instructor_manager,
              enrollment_manager, curriculum_tracker, lessons_learned
[SHARED]      theme.py
```

Key relationships: - `readiness_tracker → progress_tracker` (hard runtime dependency) - All apps `→ training_metrics` (soft aggregation) - `readiness_tracker + exam_analytics + aar_aggregator → portal` (mandatory) - `theme.py → ALL dashboards` (branding)

| | | | | **Palantir External References** | | | | Palantir Developer Community | community.palantir.com — 4,300+ topics across product Q&A, official updates, community feedback | All TMs; see source_material/palantir_community/README.md | | Palantir Public Docs | docs.palantir.com/foundry — Ontology, Data Integration, AIP, Security | All TMs; see source_material/palantir_community/PALANTIR_DOCS_REFERENCE.md | | Ontologize YouTube Channel | @Ontologize — 68 indexed videos, 24 hrs runtime; official Palantir training partner | All TMs; see source_material/ontologize_youtube/README.md | | Palantir Defense OSDK | palantir.com/defense/sdk — WFF-aligned ontology, CJADC2 compatibility | TM-40L, TM-50L | | Palantir Certifications | Builder Foundations Badge, Data Engineer Cert, App Developer Cert | TM-20, TM-30, TM-40L |

SECTION 10 — EXTERNAL REFERENCE LIBRARY: ONTOLOGIZE & PALANTIR COMMUNITY

Source: Ontologize YouTube channel (@Ontologize) — official Palantir partner training videos. **Source:** Palantir Developer Community (community.palantir.com) — product guidance, feature announcements, community Q&A. **Integration method:** (1) README.md § Supporting Reference Material — indexed with TM cross-references. (2) All 13 SELF_STUDY_ADDENDUM.md files — companion resource cross-reference to Ontologize channel. (3) Inline NOTE callouts in TM files for specific Palantir best practices (PK rules, project architecture, object type naming, Defense OSDK, Compute Modules, DDIL inference, AI-FDE). (4) Exercise packages — Build with AIP official tutorial references. **Verification:** Community content is user-generated and may reference deprecated features. "Inside the Product" posts (Palantir staff) carry higher authority. All community content should be cross-referenced against official docs and current MSS environment before incorporation.

Ontologize Channel — Coverage Summary

Topic Area	Videos	Primary TM Tracks
Getting Started / End-to-End	5	TM-10, TM-20
Data Connection	2	TM-20
Pipeline Builder	7	TM-20, TM-30
Code Repositories	3	TM-30, TM-40L, TM-40M
Ontology Design	3	TM-20, TM-30, TM-40K
Analytics (Contour/Quiver)	6	TM-10, TM-20, TM-40G
Workshop Applications	7	TM-10, TM-20, TM-30
Geospatial / Mapping	3	TM-10, TM-40A–F
AIP Logic	11	TM-40H, TM-50H
AIP Agents	7	TM-40H, TM-50H
RAG & Semantic Search	6	TM-40H, TM-40K, TM-50H
LLM & Document Processing	4	TM-40H, TM-40M
MLOps	3	TM-40M, TM-50M
Process Mining	3	TM-40D
OSDK	2	TM-40L, TM-50L

DRAFT

Topic Area	Videos	Primary TM Tracks
Data Protection	1	TM-40J, TM-40K
MCP	2	TM-40H, TM-40L
Total	**68**	**All tracks**

Palantir Community — High-Value Content Incorporated

Resource	Source	Incorporated Into	Content
Ontology & Pipeline Design Principles	community/t/5481 (Palantir staff)	TM-20, TM-30, TM-40K, Naming Standards	PK rules, project architecture taxonomy, object type naming/maturity
Defense OSDK	community/t/3561 (Palantir staff)	TM-40L, TM-50L	WFF-aligned ontology, CJADC2, Defense SDK
Why We Built It: Compute Modules	community/t/3292 (Palantir staff)	TM-40L, TM-50L	Containerized compute architecture, decision guide
DDIL / Classified Inference	community/t/6146 (community)	TM-40H, TM-50H	Local inference for AIP Logic in DDIL/classified environments
AI-FDE Core Architecture	community/t/6199 (community)	TM-50H	Modes/Skills hierarchical agent framework
Build with AIP Examples (10)	community/t/5137 (Palantir staff)	EX_40H, EX_40J, EX_40K, EX_40L	Installable tutorial templates
Certification Alignment	community/t/1043, /2789, /548	ENROLLMENT_SOP	TM→cert readiness mapping

CHANGE LOG — PALANTIR COMMUNITY & ONTOLOGIZE INTEGRATION (18 MARCH 2026)

Scope: Palantir Developer Community (community.palantir.com) scraped and indexed. Ontologize YouTube channel (68 videos) cross-referenced. Content gap analysis against all TMs completed. 10 incorporation items executed across ~30 files. All 237 PDFs force-rebuilt.

Sources: Palantir Developer Community (4 categories, ~4,300 topics), Palantir Public Docs (docs.palantir.com/foundry), Ontologize YouTube (@Ontologize, 68 videos, 24 hrs).

DRAFT

#	File(s)	Change	Source
203	source_material/palantir_community/README.md	NEW — Full community index: 6 sections (official guidance, certifications, tags, extracted content, Build with AIP, AIP agents, DDIL)	community.palantir.com scrape
204	source_material/palantir_community/PALANTIR_DOCS_REFERENCE.md	NEW — Public docs index: 4 sections (Ontology, Data Integration, AIP, Security) with TM cross-refs	docs.palantir.com/foundry
205	source_material/ontologize_youtube/README.md	NEW — 68-video index with 17 topic areas and TM track cross-reference	youtube.com/@Ontologize
206	README.md	Added Palantir Public Docs subsection, Palantir Developer Community subsection, Ontologize entry to Supporting Reference Material	All three sources
207	TM_20_BUILDER.md + CONCEPTS_GUIDE_TM20	PK Design Rules NOTE (strings only, uniqueness, no property inference)	community/t/5481
208	TM_30_ADVANCED_BUILDER.md	PK rules + Project Architecture Taxonomy (5 types) + Object Type Naming/Maturity Standards	community/t/5481
209	TM_40K_KNOWLEDGE_MANAGER.md + CONCEPTS_GUIDE_TM40K	Foundry Project Architecture Taxonomy NOTE (KM-specific)	community/t/5481
210	TM_40L_SOFTWARE_ENGINEER.md	Defense OSDK NOTE + Compute Modules architecture NOTE	community/t/3561, community/t/3292
211	TM_50L_SOFTWARE_ENGINEER_ADVANCED.md	Defense OSDK NOTE + Compute Modules architecture NOTE	community/t/3561, community/t/3292
212	TM_40H_AI_ENGINEER.md	DDIL/Classified Inference NOTE	community/t/6146
213	TM_50H_AI_ENGINEER_ADVANCED.md	DDIL/Classified Inference NOTE + AI-FDE Modes/Skills Architecture NOTE	community/t/6146, community/t/6199

DRAFT

#	File(s)	Change	Source
214	standards/NAMING_AND_GOVERNANCE_STANDARDS.md	Object Type Naming & Lifecycle Standards subsection (naming rules, maturity status, ownership)	community/t/5481
215	training_management/ENROLLMENT_SOP.md	Palantir Certification Alignment section (3 cert mappings to TM completion)	community/t/1043, /2789, /548
216	13 × SELF_STUDY_ADDENDUM.md (TM-30, TM-40G–M, TM-50G–M)	Ontologize companion resource cross-reference added after Palantir Developers channel ref	youtube.com/@Ontologize
217	exercises/EX_40H, EX_40J, EX_40K, EX_40L	Build with AIP Official Tutorials supplemental sections (5/3/3/4 tutorials respectively)	community/t/5137 + build.palantir.com
218	DEPENDENCY_MAP.md	Section 10, External Doctrinal References (Palantir), changelog entries #203–218	All sources
219	237 PDFs force-rebuilt	All source hash changes reflected; SHA256 manifest regenerated	N/A
220	pdf/.manifest.json	11 source hashes updated	N/A
221	pdf/pdf_manifest.sha256	237 entries regenerated	N/A
222	DEPENDENCY_MAP.html	Regenerated (144 nodes, 337 edges)	N/A
223	pdf/AR_350_1_ALIGNMENT_MAP.pdf	AR 350-1 alignment map (PDF): 11 regulatory areas mapped to MSS components, SAT artifact coverage table, 3 identified gaps with mitigations, compliance posture summary	AR 350-1, TR 350-70, TP 350-70-14, ADP 7-0, FM 7-0

DRAFT